

SEC P vs NP log 2026-05-17

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-17 21:20 UTC

SEC P vs NP — daily log 2026-05-17

38 cycles recorded

GCT Lie-Stabilizer Codimension Linearly Bounds ACC^0 Size

- **Verdict:** INCONCLUSIVE
- **Bridge:** Geometric Complexity Theory: Lie-algebra stabilizers of multilinear polynomial extensions in the Mulmuley-Sohoni-Bürgisser-Ikenmeyer framework $\times \text{ACC}^0[m]$ circuit size for Boolean functions (Sipser-like targets)
- **Recorded:** 2026-05-17 00:01 UTC
- **Entry ID:** 83d26c2ef069

Statement

For a Boolean function $f: \{-1, +1\}^n \rightarrow \{-1, +1\}$, let $F_f(x) = \sum_{S \subseteq [n]} \hat{f}(S) \cdot \prod_{i \in S} x_i$ be its multilinear Walsh extension over $\mathbb{Q}$, let $L(F_f) := \{A \in \mathfrak{gl}_n(\mathbb{Q}) : \sum_{i,j} A_{ij} x_j \partial_i F_f \equiv 0 \text{ in } \mathbb{Q}[x_1, \dots, x_n]\}$ be its GCT Lie stabilizer, and set $\gamma(f) := n^2 - \dim_{\mathbb{Q}} L(F_f)$. Conjecture: there is an absolute constant $C \leq 10$ such that every $\text{ACC}^0[m]$ circuit of size s computing f satisfies $\gamma(f) \geq n^2 - C \cdot s$. A single $\text{ACC}^0[m]$ circuit of size s computing some f with $\gamma(f) < n^2 - 10 \cdot s$ falsifies the conjecture.

Rationale

GCT's symmetry-characterization principle says polynomials whose stabilizer Lie algebras exceed generic dimension are atypical; we invert this to claim small ACC^0 circuits cannot accidentally synthesize multilinear extensions of anomalously high symmetry. The Lie stabilizer is GL_n -equivariant rather than truth-table-dense, so it dodges Razborov-Rudich's 'large + constructive' clause, and it is computed from the GL_n action on polynomials, not from black-box algebraic extensions — placing the invariant outside the algebrization and relativization regimes that GCT was designed to escape.

Novelty

- Judge: NOVEL over 11 arXiv hits

Top hits consulted: - [1212.5380v2] On properties of principal elements of Frobenius Lie algebras - [1606.05050v1] Proof Complexity Lower Bounds from Algebraic Circuit Complexity - [2411.11095v3] Invariant theory and coefficient algebras of Lie algebras - [1904.05483v2] Parallels Between Phase Transitions and Circuit Complexity? - [1504.04131v2] Formulas for the Walsh coefficients of smooth functions and their application to bounds on the Walsh coefficients

Empirical Test

- exit code: 1, elapsed: 0.11s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_698adb63.py", line 102, in <module>
    result = run_trial(seed)
    ~~~~~^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_698adb63.py", line 86, in run_trial
    gamma = gaussian_elimination(system)
    ~~~~~^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_698adb63.py", line 65, in gaussian_elim
    if matrix[i][i] == 0:
    ~~~~~^
```

IndexError: list index out of range

Judge reasoning

The test crashed with an `IndexError` in `gaussian_elimination` before any of the 480 trials or reference-function checks produced slack data, so the pre-registered support condition cannot be evaluated. Per Rule 2, a crash yields **INCONCLUSIVE**. | next: Fix the Gaussian elimination routine (guard against out-of-range pivot indexing and handle rectangular/short matrices) and rerun the full (n,s)-grid plus reference-function suite to obtain actual slack statistics.

2-adic Newton Slopes of DISJ Sub-blocks Scale as $\sqrt{n}$

- **Verdict:** INCONCLUSIVE
- **Bridge:** p-adic analysis (Newton polygons over $\mathbb{Q}_2$ of integer characteristic polynomials) $\times$ Randomized communication complexity of DISJOINTNESS via average-case tensor invariants on sub-blocks
- **Recorded:** 2026-05-17 00:26 UTC
- **Entry ID:** efd166108501

Statement

Let $M_n \in \{-1, +1\}^{\{2^n \times 2^n\}}$ be the signed disjointness matrix $M_n[x, y] = (-1)^{|x \cap y|}$ where x, y are interpreted as subsets of $[n]$. For each n , draw uniformly random row set R and column set C with $|R|=|C|=k=\min(2n, 40)$, form $A := M_n[R, C]$, compute its exact integer characteristic polynomial $p(t) = \sum_{i=0}^k c_i t^i$, and define the 2-adic Newton slope count $v_2(A) :=$ number of distinct slopes in the lower convex hull of $\{(i, v_2(c_i)) : c_i \neq 0\}$ where v_2 is the 2-adic valuation ($v_2(0) := +\infty$). Conjecture: $E_{R, C}[v_2(M_n^{\{R, C\}})] = \Theta(\sqrt{n})$; moreover for every $n \geq 8$ the DISJ median exceeds the median of v_2 for an i.i.d. ± 1 matrix of the same shape by a factor ≥ 1.5 , and a single (n, seed) pair where this median ratio inverts falsifies the conjecture.

Rationale

The subset semilattice underlying DISJ injects 2-adic congruence structure into every sub-block (parity of intersection size acts like a $\mathbb{Z}_2$ -valued character), so subdeterminants accumulate distinguished 2-adic valuations that random Bernoulli matrices lack — a property that should propagate, via Yao’s principle, into a lower bound on randomized communication. Newton polygons over $\mathbb{Q}_2$ are not preserved by polynomial extension to $\mathbb{F}_2[t]/(t^2-t)$, evading the algebrization barrier, and the invariant is structural rather than ‘large+useful on truth tables’, evading Razborov-Rudich. The bridge between p-adic Newton-polygon geometry and communication complexity has essentially no prior literature, while the average-to-worst-case framing ($E[v_2] \mapsto$ worst-case CC lower bound) follows the Impagliazzo-Wigderson template.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 34.76s

```
tances_tested': 5, 'conjecture_holds': False, 'counterexample': 'Ratio 1.0 < 1.5 for seed 547'}
TRIAL: {'metric_name': 'median_newton_slope_count', 'metric_value': 2, 'instances_tested': 5, 'conje
TRIAL: {'metric_name': 'median_newton_slope_count', 'metric_value': 2, 'instances_tested': 5, 'conje
TRIAL: {'metric_name': 'median_newton_slope_count', 'metric_value': 2, 'instances_tested': 5, 'conje
TRIAL: {'metric_name': 'median_newton_slope_count', 'metric_value': 2, 'instances_tested': 5, 'conje
TRIAL: {'metric_name': 'median_newton_slope_count', 'metric_value': 2, 'instances_tested': 5, 'conje
TRIAL: {'metric_name': 'median_newton_slope_count', 'metric_value': 2, 'instances_tested': 5, 'conje
TRIAL: {'metric_name': 'median_newton_slope_count', 'metric_value': 2, 'instances_tested': 5, 'conje
TRIAL: {'metric_name': 'median_newton_slope_count', 'metric_value': 2, 'instances_tested': 5, 'conje
```

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b68324ac.py", line 136, in <module>
    failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                    ~~~~~^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b68324ac.py", line 136, in <genexpr>
    failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                    ~~~~~^
```

KeyError: 'seed'

Judge reasoning

Safety rail: critic_challenge_falsified | original: Multiple trials report conjecture_holds=False with ratio $1.0 < 1.5$ (e.g., seeds 547, 593, 631, ...), satisfying the pre-registered falsification condition | next: Re-run with corrected seed-logging, stratify by $n \in \{10, 14, 18, 22\}$, and check whether the v_2 statistic saturates at small $k = \min(2n, 40)$ — perhaps a normalized slope-density (v_2/k) would better capture any DISJ-vs-random gap.

Cyclic Burnside Orbit Count Lower-Bounds AC^0 Size for PARITY

- **Verdict:** INCONCLUSIVE
- **Bridge:** Burnside-ring / equivariant orbit counting of cyclic group actions on labeled circuit DAGs (a corner of combinatorial representation theory rarely applied to circuit complexity) $\times$ AC^0 circuit size for PARITY_n, with the lifting-style intuition that gadget symmetry constrains the lifted communication matrix
- **Recorded:** 2026-05-17 00:49 UTC
- **Entry ID:** 80d12cf7a27e

Statement

Let C be an AC^0 circuit over inputs $x_0, \dots, x_{n-1}$ computing PARITY_n with n a prime ≥ 3 , $\text{depth}(C)=d$ and $\text{size}(C)=s$, and let σ be the cyclic shift $\sigma(i)=i+1 \bmod n$ acting on each gate g of C by simultaneously relabeling every input wire x_i appearing in g 's recursive expansion to $x_{\sigma(i)}$. Define $\psi(C)$ as the number of $\langle \sigma \rangle$ -orbits of gates of C under the equivalence $g \sim h$ iff some σ^k maps the lex-min canonical (op-type, sorted-children) form of g to that of h . Then for every such C computing PARITY_n we conjecture $\psi(C) \geq (1/(4d)) \cdot \log_2(s)$, refuted by any single PARITY-computing AC^0 circuit whose ratio $4 \cdot d \cdot \psi(C) / \log_2(s)$ drops below 1.

Rationale

Lifting theorems translate query lower bounds into communication lower bounds by composing the function with a symmetry-breaking gadget; dually, when the target function (PARITY) is itself shift-invariant, a small circuit must reuse few genuinely distinct sub-computations, so the Z_n -orbit count of its gates becomes a Burnside-style proxy for the work that random restrictions are usually invoked to expose. Because ψ is defined on the circuit DAG rather than on the truth table, the invariant sits outside the Razborov–Rudich natural-proofs barrier; because the construction is purely combinatorial (no polynomial extension of the Boolean ring) it does not algebrize, matching the LIFTING approach’s safe-direction profile.

Novelty

- Judge: NOVEL over 9 arXiv hits

Top hits consulted: - [2304.02770v2] Tight Correlation Bounds for Circuits Between AC0 and TC0 - [2504.19966v3] Quantum circuit lower bounds in the magic hierarchy - [1708.00951v2] Canonical Heights and Monomial Maps: On Effective Lower Bounds for Points with Dense Orbit - [1904.05483v2] Parallels Between Phase Transitions and Circuit Complexity? - [2005.06373v2] Counting Schur Rings over Cyclic Groups of Semi-prime Order

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c17bcfb6.py", line 225, in <module>
    result = run_trial(seed)
              ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c17bcfb6.py", line 180, in run_trial
    psi = compute_psi(circuit, n)
          ^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c17bcfb6.py", line 139, in compute_psi
    gate_hashes[gate] = hash_gate(gate)
                        ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c17bcfb6.py", line 126, in hash_gate
    return hash(gate)
           ^^^^^
```

TypeError: unhashable type: 'list'

Judge reasoning

The test crashed with a `TypeError (unhashable list)` in `hash_gate` before any trial completed, so no `r` values were produced and the pre-registered support condition could not be evaluated. | next: Fix the canonicalization in `hash_gate` by converting children lists to tuples (or using a frozen canonical form) before hashing, then re-run the full 2160-circuit grid.

Cauchy Mean Width of Minterm Hull Lower-Bounds Monotone k-CLIQUE

- **Verdict:** INCONCLUSIVE
- **Bridge:** Convex geometry (Cauchy mean-width / intrinsic 1-volume of polytopes in R^N , in the spirit of Schneider’s Brunn–Minkowski theory, rarely connected to circuit complexity) $\times$ Karchmer–Wigderson / monotone DNF size and formula complexity for the k-CLIQUE indicator
- **Recorded:** 2026-05-17 01:28 UTC
- **Entry ID:** 65d1ffdf3cc6

Statement

For a monotone Boolean function f on N variables, let $M(f) \subseteq \{0,1\}^N$ be its set of minterms (minimal satisfying assignments viewed as 0/1 vectors) and let $K(f) := \text{conv}(M(f) \cup \{0\}) \subseteq [0,1]^N$. Define $\mu(f) := \text{MW}(K(f))^2$, where $\text{MW}(K) = 2 \cdot \mathbb{E}_{u \sim \text{Unif}(S^{\wedge \{N-1\}})}[\max_{x \in K} \langle u, x \rangle]$ is the Cauchy mean width. We conjecture: (i) for every monotone DNF representation of f with s terms, $\mu(f) \leq C_1 \cdot \log(s+1) \cdot (\log N + 1)$; (ii) for the k -CLIQUE indicator on K_v with $k = \lfloor \log_2 v \rfloor$ (so $N = v(v-1)/2$), $\mu(f) \geq C_2 \cdot v$, for absolute constants $C_1, C_2 > 0$. A single instance with $\mu(f) > C_1 \cdot \log(s+1)(\log N + 1)$, or a k -CLIQUE indicator with $\mu < C_2 \cdot v$ at $v \leq 9$, falsifies the conjecture.

Rationale

Monotone DNFs are unions of axis-aligned subcubes, so $\text{conv}(M(f))$ is the convex hull of at most s lattice vectors of bounded coordinate sum, and its mean width should grow only logarithmically with the number of generators (a Sudakov/Talagrand-style upper bound for sparse 0/1 hulls). The k -CLIQUE minterm cloud, in contrast, lives on the $(k \text{ choose } 2)$ -th slice and is rotationally rich because v -vertex automorphisms act transitively on edges, forcing the supporting hyperplanes to spread evenly over many directions and inflating mean width linearly in v . If both bounds hold, any monotone DNF for k -CLIQUE has $s \geq 2^{\Omega(v/\log v)}$, a Razborov-flavor separation via a purely metric-geometric witness that bypasses approximator combinatorics.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 0, elapsed: 33.94s

```
me": "mean_width_squared", "metric_value": 1.136003662687513, "instances_tested": 17, "conjecture_ho
CLIQUE with v=4 has mu=1.136003662687513 < 0.5*v=2.0", "seed": 421}
TRIAL: {"metric_name": "mean_width_squared", "metric_value": 1.1857866298955364, "instances_tested"
CLIQUE with v=4 has mu=1.1857866298955364 < 0.5*v=2.0", "seed": 463}
TRIAL: {"metric_name": "mean_width_squared", "metric_value": 1.1927318541604457, "instances_tested"
CLIQUE with v=4 has mu=1.1927318541604457 < 0.5*v=2.0", "seed": 503}
TRIAL: {"metric_name": "mean_width_squared", "metric_value": 1.1396899464107255, "instances_tested"
CLIQUE with v=4 has mu=1.1396899464107255 < 0.5*v=2.0", "seed": 547}
TRIAL: {"metric_name": "mean_width_squared", "metric_value": 1.1832069622303316, "instances_tested"
CLIQUE with v=4 has mu=1.1832069622303316 < 0.5*v=2.0", "seed": 593}
TRIAL: {"metric_name": "mean_width_squared", "metric_value": 1.1742179582193337, "instances_tested"
CLIQUE with v=4 has mu=1.1742179582193337 < 0.5*v=2.0", "seed": 631}
TRIAL: {"metric_name": "mean_width_squared", "metric_value": 1.1635078759135067, "instances_tested"
CLIQUE with v=4 has mu=1.1635078759135067 < 0.5*v=2.0", "seed": 677}
TRIAL: {"metric_name": "mean_width_squared", "metric_value": 1.1855075477194135, "instances_tested"
CLIQUE with v=4 has mu=1.1855075477194135 < 0.5*v=2.0", "seed": 727}
TRIAL: {"metric_name": "mean_width_squared", "metric_value": 1.172332125064841, "instances_tested":
CLIQUE with v=4 has mu=1.172332125064841 < 0.5*v=2.0", "seed": 773}
TRIAL: {"metric_name": "
```

Judge reasoning

Safety rail: critic_challenge_falsified | original: Every k -CLIQUE trial at $v=4$ yields $\mu \approx 1.17$, well below the falsification threshold $0.25 \cdot v = 1.0$... actually $0.25 \cdot 4 = 1.0$ so this passes the $0.25 \cdot v$ floor | next: Rescale the lower bound to $\mu(f) \geq C_2 \cdot v / \log v$ or test the conjecture with normalized mean width $\text{MW}(K(f)) / \sqrt{N}$ to remove metric-saturation artifacts at small v .

Costas Displacement Coincidence Lower-Bounds AC^0 Size for PARITY

- **Verdict:** INCONCLUSIVE
- **Bridge:** Costas array / Welch–Costas displacement combinatorics (Golomb–Costas radar arrays and B₂/Sidon-deficiency theory of integer sets) — a corner of additive combinatorics developed for radar/sonar that has essentially no presence in circuit complexity literature × AC^0 depth-d circuit size for PARITY_n (Hastad-style lower bounds, viewed through the lens of bottom-gate support designs)
- **Recorded:** 2026-05-17 02:01 UTC
- **Entry ID:** 6bb89da9ec2e

Statement

Let C be a depth- d AC^0 circuit on n inputs with bottom-level (depth-1) AND/OR gates having literal-supports $S_1, \dots, S_m \subseteq [n]$. Define the Costas displacement defect $\kappa(C) = \log_2(1 + \max_{\delta \in \{1, \dots, n-1\}} |\sum_{i=1}^m |\{(a,b) \in S_i \times S_i : a > b \text{ and } a-b \equiv \delta \pmod{n}\}| |)$. We conjecture that for every C that computes PARITY_n exactly, $\kappa(C) \geq (1/4) \cdot n^{1/(d-1)}$, and consequently $\text{size}(C) \geq 2^{\Omega(\kappa(C))}$. A single depth- d AC^0 circuit that computes PARITY_n while having $\kappa(C) < (1/4) \cdot n^{1/(d-1)}$ refutes the conjecture.

Rationale

PARITY is translation-invariant on the Boolean cube, so any AC^0 circuit computing it must spread its bottom-level supports densely across all cyclic displacements — exactly the property that Costas/Welch arrays minimize. Replacing Hastad’s random restriction by a Costas-style design-theoretic lower bound turns AC^0 PARITY bounds into a question about Sidon-deficiency of support multisets. The invariant depends only on the circuit’s bottom-support combinatorics (not the function’s truth table), so it sidesteps Natural Proofs, and uses no ring operations, sidestepping algebrization.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 124, elapsed: 240.18s

TIMEOUT after 240s

Judge reasoning

The test timed out after 240s (returncode 124) without producing a RESULT line, so neither the support nor falsification condition could be evaluated. | next: Reduce the parameter sweep (e.g., cap n at 20 and $d=2$) or parallelize per-seed evaluation to produce data within the timeout.

Vertex Star-Discrepancy Equals Spectral Norm for XOR Comm

- **Verdict:** INCONCLUSIVE
- **Bridge:** Geometric discrepancy theory (vertex-restricted L_∞ star discrepancy of $\{0,1\}^n$ point sets, in the Chazelle–Matousek tradition of axis-parallel box discrepancy) × Randomized two-party communication complexity of XOR-functions $f(x,y)=g(x \oplus y)$ accessed through the discrepancy-method lower bound $\text{disc}(M_f)=\|\hat{g}\|_\infty$
- **Recorded:** 2026-05-17 02:29 UTC
- **Entry ID:** d27e7e45b29b

Statement

For any non-constant $g: \{0,1\}^n \rightarrow \{-1,+1\}$, let $P_g = \{z \in \{0,1\}^n : g(z) = -1\}$ and define the lattice star discrepancy $D^*_{lat}(g) := \max\{a \in \{0,1\}^n \mid |P_g \cap [0,a] \square| / 2^n - (|P_g|/2^n)^{1/2} \{popcount(a)\}/2^n\}$, where $[0,a] \square = \{z \in \{0,1\}^n : z \leq a \text{ coordinatewise}\}$; let $\|\hat{g}\|_\infty := \max\{S \subseteq [n] \mid |\hat{g}(S)|\}$ be the spectral norm. Conjecture: there exist absolute constants $c_1=1/8$ and $c_2=8$ such that for every $n \geq 4$ and every non-constant g , $c_1 \cdot \|\hat{g}\|_\infty \leq D^*_{lat}(g) \leq c_2 \cdot \|\hat{g}\|_\infty \cdot \log_2(n+1)$. A single (n,g) pair violating either inequality refutes the conjecture.

Rationale

The discrepancy lower bound on randomized XOR-communication is exactly $\log(1/\|\hat{g}\|_\infty)$, yet $\|\hat{g}\|_\infty$ is a global Fourier quantity with no geometric face. Vertex star discrepancy is the canonical Chazelle–Matousek irregularity measure of a point set inside $[0,1]^n$; restricting test boxes to lattice corners makes it computable in $O(n \cdot 2^n)$ via the zeta transform. If $D^*_{lat} \approx \|\hat{g}\|_\infty$, this gives the first concrete geometric (axis-parallel box) certificate of XOR-communication hardness, opening a Banaszczyk/Kashin-style geometric attack on unsaturated $R(\text{DISJ})$ -like bounds.

Novelty

- Judge: NOVEL over 2 arXiv hits

Top hits consulted: - [s2:1704.02537] Dual polynomials and communication complexity of XOR functions - [s2:19311e592db3bd54ad8685ae0d27b815c87be] Dual polynomials and communication complexity of XOR functions

Empirical Test

- exit code: 124, elapsed: 240.03s

TIMEOUT after 240s

Judge reasoning

The test timed out after 240s (returncode=124) without producing a RESULT line or any trial data, so neither the support nor falsification condition can be evaluated. | next: Reduce the maximum n or restrict the enumeration over $a \in \{0,1\}^n$ (e.g., sample or use a smarter combinatorial bound) so the 210 trials complete within the time budget.

Operator-SoS 4-Trace Gap Lower-Bounds Blocks-Order Read-Twice BP for IP_2

- **Verdict:** INCONCLUSIVE
- **Bridge:** Sum-of-Squares / Lasserre hierarchy on operator-valued layer-product tensor moments (Putinar–Helton-style non-commutative SoS pseudo-trace invariants applied to layer-difference matrices of branching programs — a corner combining moment-SoS literature with BP structural invariants, with <5 papers on the BP side) × Read-twice oblivious branching program size for IP_2 (the open BP_READTWICE separation of Borodin–Razborov–Smolensky 1993)
- **Recorded:** 2026-05-17 02:59 UTC
- **Entry ID:** 0ecd5f06ea81

Statement

Let P be a read-twice oblivious BP on $2n$ Boolean variables $(x_1, \dots, x_n, y_1, \dots, y_n)$ with width w and length $4n$, layer transition matrices $M_j(0), M_j(1) \in \{0,1\}^{w \times w}$, and layer-difference matrices

$N_j := M_j(1) - M_j(0)$; for each variable v with its two read-layers (a_v, b_v) form the paired operator $P_v := N_{\{a_v\}} N_{\{b_v\}} \in \mathbb{R}^{w \times w}$, and define the operator-SoS 4-trace gap $\rho(P) := \log_2 \max_{\{u \neq v\}} |\langle P_u, P_v \rangle_F| / w^2$, where $\langle A, B \rangle_F = \text{tr}(A B^T)$. CONJECTURE: (Upper anchor, trivial via SoS-Cauchy-Schwarz) $\rho(P) \leq 2 \cdot \log_2 \text{width}(P) + 2$ for every read-twice oblivious BP P ; (Lower bound, non-trivial) for every BLOCKS-ORDERED read-twice oblivious BP P (all x -layers preceding all y -layers in each of the two reads) that computes $\text{IP}_2(x, y) = \bigoplus_i x_i y_i$ with bias $\geq 1/2 + n^{-2}$, $\rho(P) \geq n/16 - 4 \cdot \log_2 n$. Equivalently, any such blocks-ordered read-twice BP for IP_2 has width $\geq 2^{\{n/40\}}$ for all sufficiently large n ; a single small-width blocks-ordered read-twice BP for IP_2 with $\rho(P) < n/16 - 4 \cdot \log_2 n$ falsifies the conjecture.

Rationale

The paired operator $P_v = N_{\{a_v\}} N_{\{b_v\}}$ packages the two reads of variable v into a single non-commutative SoS-style moment, and the Frobenius pairing $\langle P_u, P_v \rangle_F$ is exactly the kind of trace identity that survives Lasserre relaxation at level 2 yet remains computable directly from the BP. For IP_2 the algebraic obstruction that forces exponential read-once size — namely that x and y blocks cannot be summarized in fewer than 2^n states without bilinear cancellation — should reappear in the read-twice setting as a forced spread of mass across many variable-pairs in the Gram matrix of $\{P_v\}$, producing a heavy off-diagonal entry. Because ρ depends on the specific BP layer matrices (not the truth table) it cannot collapse to a function-symmetric invariant, side-stepping the natural-proofs trap that killed the prior Lasserre-rank attempt in this approach.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8fle5764.py", line 113, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8fle5764.py", line 69, in run_trial
    N = generate_bp(w, n, seed)
        ^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8fle5764.py", line 44, in generate_bp
    N = [matrix_subtract(M[j], M[j-1]) if j > 0 else M[j] for j in range(2*n)]
         ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8fle5764.py", line 28, in matrix_subtra
    w = len(A)
        ^^^^^
```

TypeError: object of type 'int' has no len()

Judge reasoning

The test crashed with a `TypeError` in `matrix_subtract` before producing any data, so neither the upper-anchor nor the IP_2 lower-bound criterion was evaluated. No `RESULT` line was emitted and the pre-registered support condition cannot be assessed. | next: Fix the matrix representation bug in `generate_bp/matrix_subtract` (ensure `M[j]` is a 2D list/array of shape $w \times w$, not an `int`) and re-run the 540 random instances plus the 4 explicit IP_2 BPs.

Cross-Side Fourier Symmetric-Difference Mass Lower-Bounds DISJ CC_R

- **Verdict:** BARRIER_HIT
- **Bridge:** Fourier analysis of Boolean functions (Bonami-Beckner / KKL-style level-weighting on the cube, with bipartite cross-coordinate noise operators rarely used in CC) × Randomized two-party communication complexity of DISJOINTNESS (CC_R) and other partial functions on $N=2^n \times 2^n$ matrices
- **Recorded:** 2026-05-17 03:06 UTC
- **Entry ID:** fba37ecf9a43

Statement

For a Boolean matrix $M \in \{0,1\}^{2n \times 2^n}$ viewed as a function $f_M : \{0,1\}^{2n} \rightarrow \{0,1\}$ with first n bits the x -input and last n bits the y -input, and for $T \subseteq [2n]$ let $T_x = T \cap [n]$ and $T_y = \{i \in [n] : i+n \in T\}$, define $\delta(T) := |T_x \Delta T_y|$ and the cross-asymmetry invariant $\Psi(f_M) := (\sum_T f_M \hat{T})^2 \cdot 2^{\delta(T)} / (\sum_T f_M \hat{T})^2$. Conjecture: there exists an absolute constant $c \geq 1/8$ such that for every Boolean matrix M , $CC_R^{1/3}(M) \geq c \cdot \log_2 \Psi(f_M)$, and moreover $\Psi(f_{\text{DISJ}_n}) = (7/6)^n$ exactly (so $\log_2 \Psi = n \cdot \log_2(7/6) \approx 0.222 n$ and $CC_R(\text{DISJ}_n) = \Omega(n)$ follows). A single Boolean matrix M with a verified protocol of cost B and $\log_2 \Psi(f_M) > 8 \cdot B$ refutes the conjecture.

Rationale

Standard Fourier ℓ_1 / spectral norm lower bounds for CC are oblivious to the bipartition structure of the input, while CC_R cares essentially about which Fourier modes ‘mix’ the two parties. Weighting each Fourier mass by $2^{|T_x \Delta T_y|}$ amplifies modes that genuinely couple x - and y -coordinates asymmetrically — EQ-type matrices live on the diagonal $T_x = T_y$ and get $\Psi = 1$, while DISJ’s product-over-coordinates structure forces a multiplicative gain of $7/6$ per coordinate, giving the right Razborov-type $\Omega(n)$. The hypothesis is therefore a Fourier-analytic ‘cross-side’ refinement of the standard discrepancy/spectral-norm method, in the spirit of KKL-style coordinate-weighted Fourier inequalities.

Novelty

- Judge: “ over 0 arXiv hits

Empirical Test

- exit code: -1, elapsed: 0.00s

(no output)

Judge reasoning

Rejected by NATURAL_PROOFS barrier

q-Major Cancellation Gap Separates Permanent from Determinant Supports

- **Verdict:** INCONCLUSIVE
- **Bridge:** Mahonian statistics on the symmetric group (major index, q -factorials, RSK cocharge) × Padded permanent/determinant comparison on shared 0/1 support matrices (GCT det-vs-perm flavor)
- **Recorded:** 2026-05-17 03:32 UTC
- **Entry ID:** 796531c3b7fa

Statement

For an $n \times n$ matrix M in $\{0,1\}^{\{n \times n\}}$, let $T_M = \{\sigma \text{ in } S_n : M[i,\sigma(i)] = 1 \text{ for all } i\}$ and define the q -major permanent and signed q -major determinant on the support T_M by $\text{perm}_q(M) := \sum_{\sigma \text{ in } T_M} q^{\{\text{maj}(\sigma)\}}$ and $\det_q(M) := \sum_{\sigma \text{ in } T_M} \text{sign}(\sigma) * q^{\{\text{maj}(\sigma)\}}$, where $\text{maj}(\sigma)$ is the major index (sum of descent positions). Conjecture: for every $n \geq 3$ and every 0/1 matrix M with $|T_M| \geq 2$, the $q=2$ cancellation ratio $R_2(M) := \text{perm}_2(M) / \max(1, |\det_2(M)|)$ satisfies $R_2(M) \geq \sqrt{|T_M|} / (2^n)$. Equivalently, the $\text{sign}(\sigma)$ -weighted Mahonian sum on any 0/1-support exhibits at least square-root-of-support cancellation against the unsigned version, up to a linear factor in n .

Rationale

Permanent vs determinant in GCT differs precisely by replacing $\text{sign}(\sigma)$ with 1 inside the S_n -sum; weighting both sums by $q^{\{\text{maj}(\sigma)\}}$ (a Schur-positive Hecke-algebra principal specialization) refracts that difference through irreducible S_n -character spectra, and at $q=2$ one already sees $[n]_2! = \prod(2^k - 1)$ for perm_n but a much smaller signed sum for det_n . A square-root cancellation lower bound on every common support T_M would certify that determinantal coefficient sequences on permanental supports always look ‘random-like’ relative to their unsigned partners, giving a Mahonian witness for the per-vs-det gap that is preserved under support-preserving substitutions.

Novelty

- Judge: NOVEL over 9 arXiv hits

Top hits consulted: - [1503.00822v2] Product Ranks of the 3×3 Determinant and Permanent
- [2302.11221v6] A q-analog of certain symmetric functions and one of its specializations -
[1302.4419v3] The determinant of the Malliavin matrix and the determinant of the covariance
matrix for multiple integrals - [0810.2731v3] Fix-Euler-Mahonian statistics on wreath products -
[1101.4332v2] Mahonian Pairs

Empirical Test

- exit code: 1, elapsed: 1.37s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c5e11f1c.py", line 51, in <module>
    result = run_trial(seed)
            ^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c5e11f1c.py", line 33, in run_trial
    perm_q = sum(2 ** sum(i for i, x in enumerate(sigma) if sigma[i] > sigma[j])) for sigma in permutat
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c5e11f1c.py", line 33, in <genexpr>
    perm_q = sum(2 ** sum(i for i, x in enumerate(sigma) if sigma[i] > sigma[j])) for sigma in permutations
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c5e11f1c.py", line 33, in <genexpr>
    perm_q = sum(2 ** sum(i for i, x in enumerate(sigma) if sigma[i] > sigma[j])) for sigma in permutation
```

```
NameError: name 'j' is not defined
```

Judge reasoning

The test crashed with a `NameError ('j' undefined)` in the major-index computation before any trial completed, so no data was produced to evaluate the pre-registered criterion. | next: Fix the `maj(sigma)` computation to correctly count descent positions (i where `sigma[i] > sigma[i+1]`) and rerun the 240-trial sweep.

Immanant Positivity Width Bounds Monotone KW Depth for Perfect Matching

- **Verdict:** INCONCLUSIVE
- **Bridge:** Stembridge-Stanley immanant theory: the λ -immanants $\text{imm}_\lambda(S) = \sum_\sigma \chi_\lambda(\sigma) \prod_i S[i, \sigma(i)]$ organise matchings of a 0/1 matrix into Schur-Weyl S_n -isotypic cells (a corner of symmetric-group representation theory with very few papers connecting it to monotone circuit lower bounds outside of GCT speculation) $\times$ Monotone Karchmer-Wigderson tree depth $d_{\text{KW}}(S)$ at a 1-input S of the bipartite perfect-matching function PM_n , equivalent by KW to $\log_2$ of the local monotone formula leaf size restricted to matchings inside S vs. small bipartite vertex covers
- **Recorded:** 2026-05-17 04:06 UTC
- **Entry ID:** 15b45d9e4966

Statement

For $3 \leq n \leq 6$ and any $n \times n$ 0/1 matrix S with $\text{perm}_n(S) \geq 1$, let $\Sigma(S) := \{\sigma \in S_n : S[i, \sigma(i)] = 1 \forall i\}$ and define the immanant positivity width $w(S) := \#\{\lambda \vdash n : \text{imm}_\lambda(S) > 0\}$. Let $d_{\text{KW}}(S)$ be the minimum depth of any KW protocol that on Alice-input $\tau \in \Sigma(S)$ and Bob-input C , an $(n-1)$ -vertex cover of the bipartite complement of some 0-input S' , outputs an edge $(i, j) \in \tau \cap C$. Conjecture: $d_{\text{KW}}(S) \geq \lceil \log_2 w(S) \rceil$ for every such S , with equality only when $\Sigma(S)$ is RSK-shape-monochromatic; in particular $w(I_n) = p(n)$, forcing $d_{\text{KW}}(I_n) \geq \lceil \log_2 p(n) \rceil = \Theta(\sqrt{n})$, a representation-theoretic obstruction in the spirit of GCT's perm/det separation.

Rationale

Each rectangle of a KW protocol pools matchings that all hit a common edge of Bob's cover; we conjecture this pooling is incompatible with carrying more than one irreducible χ_λ -component, so the number of non-vanishing λ -immanants is a Schur-Weyl obstruction to small KW trees. This mirrors the GCT philosophy that perm_n 's orbit closure is Schur-rich while det_n 's is sign-isotypic: a small substitution-stable Schur-width invariant would translate directly into a perm-vs-det monotone gap. The link of immanants to KW depth is, to our knowledge, untouched in the literature.

Novelty

- Judge: NOVEL over 9 arXiv hits

Top hits consulted: - [2107.05128v2] Karchmer-Wigderson Games for Hazard-free Computation - [2002.07444v1] Multiparty Karchmer-Wigderson Games and Threshold Circuits - [2003.09602v2] The Number of Perfect Matchings in Möbius Ladders and Prisms - [2304.05285v1] Hook-Shape Immanant Characters from Stanley-Stembridge Characters - [1801.08709v2] Adaptive Lower Bound for Testing Monotonicity on the Line

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_70051383.py", line 178, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_70051383.py", line 132, in run_trial
    d_KW = kw_depth(matrix, covers)
           ^^^^^^^^^^^^^^^^^^^^^^^
```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_70051383.py", line 104, in kw_depth
subsets[cover].append(sigma)

~~~~~^~~~~~

TypeError: unhashable type: 'set'

## Judge reasoning

The test crashed with a `TypeError` ('unhashable type: set') in `kw_depth` before producing any data, so no metrics were computed and the pre-registered support condition cannot be evaluated. | next: Fix the `kw_depth` implementation by using a hashable key (e.g., `frozenset`) for cover keys in the `subsets` dict, then rerun the 360-matrix sweep.

## Curto-Fialkow Hankel Defect Separates Read-Twice IP<sub>2</sub> BPs

- **Verdict:** INCONCLUSIVE
- **Bridge:** Curto-Fialkow truncated moment problem theory (Hankel positivity and flat-extension rank, the real-algebraic-geometry foundation of the Lasserre SoS hierarchy; Curto-Fialkow 1996, Lasserre 2001) applied to layer-conditional empirical moments of branching programs  $\times$  Read-twice oblivious branching program size for IP<sub>2</sub> on  $2n$  variables in adversarial order  $x_1, \dots, x_n, y_1, \dots, y_n$  (Borodin-Razborov-Smolensky 1993 open separation)
- **Recorded:** 2026-05-17 04:32 UTC
- **Entry ID:** 9f09553bfd37

## Statement

Let  $P$  be a read-twice oblivious BP with state set  $Q$  ( $|Q|=s$ ) computing a Boolean function on  $N=2n$  variables, and let  $q(z, \square)$  denote its state on input  $z$  after  $\square$  steps. For each state  $q \in Q$  and each layer  $\square \in \{1, \dots, 2N\}$ , define the centered conditional moment sequence  $m^{\{q, \square\}}(P) = 2^{-N} \sum_z \square[q(z, \square) = q] \cdot (IP_2(z) - 1/2)^k$  for  $k=0, \dots, \lfloor N/4 \rfloor$ , and let  $H^{\{q, \square\}}(P)$  be the  $(d+1) \times (d+1)$  Hankel matrix with  $d = \lfloor N/8 \rfloor$  and entries  $m^{\{q, \square\}}(P)^{\{i+j\}}$ . Define the positivity defect  $\delta(P) = \max_{\{q, \square\}} \#\{\text{negative eigenvalues of } H^{\{q, \square\}}(P)\}$ . Conjecture: (i) for every read-twice oblivious BP  $P$  of size  $s$ ,  $\delta(P) \leq 8 \cdot \lceil \log_2(s+1) \rceil$ ; (ii) every read-twice oblivious BP  $P$  that exactly computes IP<sub>2</sub> in the adversarial schedule satisfies  $\delta(P) \geq \lceil \sqrt{N/2} \rceil$ .

## Rationale

Curto-Fialkow theory equates degree- $2d$  SoS feasibility with the existence of flat positive extensions of truncated Hankel sequences, so a small BP forces low-rank flat extensions of its layer-conditional empirical moments — a structural constraint never previously exploited for BP lower bounds. IP<sub>2</sub>'s bilinear  $\oplus$ -structure should force every conditional sub-population reaching a given state at a given layer to retain non-trivial bilinear correlation between the unread  $x$ -bits and the moments, obstructing flat positive Hankel extension and producing many negative eigenvalues. The conjectured gap  $\log(s)$  vs  $\sqrt{N}$  would yield  $s \geq 2^{\Omega(\sqrt{N})}$  for read-twice IP<sub>2</sub>, attacking the BRS-1993 problem via the SoS-moment side rather than via communication or partition arguments.

## Novelty

- Judge: NOVEL over 0 arXiv hits

## Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8925cd5b.py", line 155, in <module>
    result = run_trial(int(seed))
    ~~~~~^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8925cd5b.py", line 104, in run_trial
 P, Q, N = generate_random_bp(n, w, seed)
    ~~~~~^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8925cd5b.py", line 64, in generate_random_bp
    P[(q, l)][k] += 1
    ~~~~~^

```

TypeError: list indices must be integers or slices, not float

## Judge reasoning

The test crashed with a `TypeError` in `generate_random_bp` before producing any data, so neither the support nor falsification condition could be evaluated. | next: Fix the indexing bug in `generate_random_bp` (cast the index `k` to `int`) and rerun the pre-registered sweep.

---

## Hodge-Cheeger Product Lower-Bounds Tseitin Resolution Length

- **Verdict:** INCONCLUSIVE
- **Bridge:** Combinatorial Hodge theory on graphs (pairing of harmonic 1-cochain dimension  $b_1$  with vertex-Laplacian algebraic connectivity, in the Lim 2020 / Jiang-Lim-Yao-Ye framework used in TDA but essentially unused in proof complexity)  $\times$  Tree-like Resolution refutation length of Tseitin XOR formulas  $T(G, \omega)$  on bounded-degree graphs  $G$  with odd charge  $\omega$
- **Recorded:** 2026-05-17 05:07 UTC
- **Entry ID:** b60e3f3f26e1

## Statement

Let  $G=(V,E)$  be a connected graph with max degree  $d_{\max} \leq 6$  and  $|V|=n$ , and let  $\omega:V \rightarrow \mathbb{F}_2$  satisfy  $\sum_v \omega_v = 1 \pmod{2}$ . Define  $b_1(G) := |E|-|V|+1$  (the first combinatorial Betti number, i.e. cycle-space dimension over  $\mathbb{F}_2$ ), let  $\lambda_2(L_G)$  denote the second-smallest eigenvalue of the combinatorial vertex Laplacian  $L_G=D-A$ , and set the Hodge-Cheeger invariant  $\nu(G) := \lambda_2(L_G) * b_1(G) / d_{\max}$ . Conjecture: there is an absolute constant  $c \geq 0.5$  such that every tree-like Resolution refutation of  $T(G, \omega)$  has length at least  $2^{\lfloor c\nu(G) \rfloor}$ ; *a single  $(G, \omega)$  with  $\nu(G) \geq 5$  and a Resolution refutation of length  $< 2^{\lfloor 0.5\nu(G) \rfloor}$  falsifies the conjecture.*

## Rationale

Tseitin hardness needs two ingredients: enough independent cycles to encode unsatisfiable XOR obstructions (measured by  $b_1$ ) and enough spectral mixing to prevent localized cancellation (measured by  $\lambda_2$ ); combinatorial Hodge theory packages exactly this pairing via the orthogonal decomposition of edge flows into gradients and harmonic 1-cochains. Bridges/paths zero out  $b_1$ ; bottlenecked graphs zero out  $\lambda_2$ ; only graphs that are simultaneously cycle-rich and spectrally expanding (the BSW regime) yield large  $\nu$ , matching the empirical hard-Tseitin profile. The framing is novel for proof complexity ( $<5$  papers connect Hodge theory to Resolution) yet entirely poly-time computable via a single numpy eigendecomposition.

## Novelty

- Judge: NOVEL over 10 arXiv hits

Top hits consulted: - [2209.05839v3] On bounded depth proofs for Tseitin formulas on the grid; revisited - [2103.09609v1] Characterizing Tseitin-formulas with short regular resolution refutations

- [0906.0693v3] An improved lower bound on the counterfeit coins problem - [1301.2045v3] Integral-valued polynomials over the set of algebraic integers of bounded degree - [1606.05050v1] Proof Complexity Lower Bounds from Algebraic Circuit Complexity

## Empirical Test

- exit code: 124, elapsed: 240.10s

TIMEOUT after 240s

## Judge reasoning

The test timed out after 240s (returncode 124) before producing any RESULT line or data, so the pre-registered support condition cannot be evaluated. | next: Reduce instance count or graph sizes (e.g., shrink to 3 categories  $\times$  3 sizes  $\times$  10 seeds) and/or impose a per-instance Resolution-search timeout so the suite completes within the wall-clock budget.

---

## Sandpile Group Order Lower-Bounds Monotone KW Depth

- **Verdict:** INCONCLUSIVE
- **Bridge:** Abelian sandpile / chip-firing group theory (Dhar 1990, Bjorner-Lovasz-Shor 1991; the critical/sandpile group  $K(G)$  = coker of the reduced graph Laplacian,  $|K(G)|$  = number of spanning trees by Kirchhoff). This corner of algebraic combinatorics / statistical mechanics has essentially no presence in monotone circuit lower-bound literature.  $\times$  Monotone Karchmer-Wigderson protocol depth  $D_{KW}(f) = \log_2$  of minimum monotone formula leaf size for a monotone Boolean function  $f$ , accessed via the bipartite KW relation (minterm, maxterm)  $\rightarrow$  intersection coordinate.
- **Recorded:** 2026-05-17 05:45 UTC
- **Entry ID:** f75d64a69735

## Statement

For every nonconstant monotone Boolean  $f: \{0,1\}^n \rightarrow \{0,1\}$  let  $M(f)$  and  $K(f)$  be its minterm and maxterm antichains (as subsets of  $[n]$ ). Form the bipartite multigraph  $G(f)$  on vertex set  $M(f) \sqcup K(f)$  with an edge of multiplicity  $|m \cap k|$  between each  $m \in M(f)$  and  $k \in K(f)$ , add a ground vertex  $g$  joined by a single edge to every  $k \in K(f)$ , let  $\tau(f) =$  number of spanning trees of  $(G(f) \cup \{g\})$  computed via the matrix-tree theorem on the reduced Laplacian, and put  $\sigma(f) = \log_2(1 + \tau(f))$ . Then  $\sigma(f) \leq D_{KW}(f) \cdot (|M(f)| + |K(f)| + 1) \cdot \log_2(n+2)$ , with the constant 1 in front being the conjectured tight prefactor. A single monotone  $f$  exhibiting  $\sigma(f)$  strictly above this bound (using any verified upper bound on  $D_{KW}(f)$  from an explicit protocol) falsifies the conjecture.

## Rationale

The sandpile group of the KW bipartite intersection multigraph counts independent ‘cycles of certificates’ that any KW protocol must resolve; each protocol step (a coordinate query at an internal node) can collapse at most  $O((|M| + |K|) \log n)$  such independent cycles by a Cheeger-style averaging on the Laplacian, suggesting depth lower-bounds via  $\log |K(G)|$ . Because  $\tau(f)$  is determined by the BOTH-sided minterm/maxterm relation rather than a single-number truth-table statistic, the invariant is relational and exponential-time in the truth table, so it sidesteps the Razborov-Rudich natural-proofs barrier and the ring-only algebrization trap.

## Novelty

- Judge: NOVEL over 9 arXiv hits

Top hits consulted: - [2107.05128v2] Karchmer-Wigderson Games for Hazard-free Computation - [1012.2478v2] Packing spanning trees and the  $k$ -tree protocol - [0904.3087v4] Distributed Maintenance of Anytime Available Spanning Trees in Dynamic Networks - [1606.05050v1] Proof Complexity Lower Bounds from Algebraic Circuit Complexity - [1904.05483v2] Parallels Between Phase Transitions and Circuit Complexity?

## Empirical Test

- exit code: 124, elapsed: 240.10s

TIMEOUT after 240s

## Judge reasoning

The test timed out after 240s (returncode 124) without producing a RESULT line, so the pre-registered support condition cannot be evaluated and the critic challenge stands. | next: Optimize the test harness (e.g., cache D\_KW upper bounds, reduce seed count or cap n at 6) so all 35 instances complete within the time budget.

---

## Conway Temperature Bounds Tree-Resolution Width for 3-SAT

- **Verdict:** INCONCLUSIVE
- **Bridge:** Conway-Berlekamp combinatorial game theory: left/right stops and temperature of finite alternating scoring games (Conway 1976; Berlekamp-Wolfe 1994; Siegel 2013). This corner of CGT has essentially no presence in proof complexity outside Beck’s positional-game tradition; a focused arXiv search for ‘Conway temperature’ AND (‘resolution’ OR ‘CNF’ OR ‘proof complexity’) returns no hits. × Tree-like Resolution refutation width  $w(F)$  and size  $s(F)$  for unsatisfiable 3-CNFs, viewed in the Cook-Reckhow-Krajicek bounded-arithmetic framework where  $\neg F$  is a  $\Sigma^b_0$  statement whose  $V^{0/S}_{1,2}$  witnessing complexity is tied by Ben-Sasson-Wigderson to Resolution width and by Pudlak-Impagliazzo to the Prover-Adversary game.
- **Recorded:** 2026-05-17 06:17 UTC
- **Entry ID:** 18df83a46e71

## Statement

For unsat 3-CNF  $F$  on  $n \leq 14$  variables, define the alternating integer-scoring game  $G(F)$ : players  $A$  (Maximizer) and  $B$  (Minimizer) alternate; on each turn the active player picks one unassigned variable and freely assigns a value; when all variables are assigned the score equals the number of falsified clauses. Compute the Conway left stop  $L(F)$  (Maximizer to move) and right stop  $R(F)$  (Minimizer to move) by memoized minimax on the full  $2^n$  game tree, and let  $t(F) := (L(F) - R(F))/2$  be the (degree-0) Conway temperature. Conjecture: (i)  $t(F) \leq C_1 \cdot w(F)$  for every unsat 3-CNF  $F$ , where  $w(F)$  is the tree-Resolution refutation width, and (ii)  $t(F) \geq C_2 \cdot n$  for Tseitin formulas  $T(G, \omega)$  on 3-regular expander graphs  $G$  with  $v \leq 10$  vertices; a single instance violating either bound refutes the conjecture (defaults  $C_1 = 5$ ,  $C_2 = 0.3$ ).

## Rationale

Tree-Resolution width is controlled by adversarial variable choice in the Prover-Adversary game, while the Conway temperature measures opposite-sign oscillation in optimal play of a scoring game — a structurally distinct invariant never (to our knowledge) computed on SAT-derived games. Tseitin formulas on expanders have  $\Omega(n)$  resolution width by parity-flow, and the parity structure should also force balanced left/right moves at every vertex-clause group, yielding high temperature. If the bridge holds it gives a witnessing-style certificate of width inside the bounded-arithmetic  $\Sigma^b_0$  Cook-Reckhow framework.

## Novelty

- Judge: NOVEL over 6 arXiv hits

Top hits consulted: - [1709.03479v2] Conway’s potential function via the Gassner representation - [2203.14669v1] Dynamic Structure in Four-strategy Game: Theory and Experiment - [2207.14140v1] Playing a 2D Game Indefinitely using NEAT and Reinforcement Learning - [2501.08313v1] MiniMax-01: Scaling Foundation Models with Lightning Attention - [2506.13585v1] MiniMax-M1: Scaling Test-Time Compute Efficiently with Lightning Attention

## Empirical Test

- exit code: 0, elapsed: 4.20s

```
: 1, 'conjecture_holds': False, 'counterexample': 'Random 3-CNF with n=10 violated $t(F) \leq 5 \cdot w^*(F)$: t=6.0, w*=True'}
TRIAL: {'metric_name': 'temperature_to_width_ratio', 'metric_value': 0, 'instances_tested': 2, 'conjecture_holds': False, 'counterexample': 'Random 3-CNF with n=10 violated $t(F) \leq 5 \cdot w^*(F)$: t=6.5, w*=True'}
CNF with n=10 violated $t(F) \leq 5 \cdot w^*(F)$: t=6.5, w*=True'}
TRIAL: {'metric_name': 'temperature_to_width_ratio', 'metric_value': 0, 'instances_tested': 1, 'conjecture_holds': False, 'counterexample': 'Random 3-CNF with n=10 violated $t(F) \leq 5 \cdot w^*(F)$: t=6.5, w*=True'}
CNF with n=10 violated $t(F) \leq 5 \cdot w^*(F)$: t=6.5, w*=True'}
TRIAL: {'metric_name': 'temperature_to_width_ratio', 'metric_value': 0, 'instances_tested': 2, 'conjecture_holds': False, 'counterexample': 'Random 3-CNF with n=10 violated $t(F) \leq 5 \cdot w^*(F)$: t=6.5, w*=True'}
CNF with n=10 violated $t(F) \leq 5 \cdot w^*(F)$: t=6.5, w*=True'}
TRIAL: {'metric_name': 'temperature_to_width_ratio', 'metric_value': 0, 'instances_tested': 2, 'conjecture_holds': False, 'counterexample': 'Random 3-CNF with n=14 violated $t(F) \leq 5 \cdot w^*(F)$: t=9.5, w*=True'}
CNF with n=14 violated $t(F) \leq 5 \cdot w^*(F)$: t=9.5, w*=True'}
TRIAL: {'metric_name': 'temperature_to_width_ratio', 'metric_value': 0, 'instances_tested': 2, 'conjecture_holds': False, 'counterexample': 'Random 3-CNF with n=10 violated $t(F) \leq 5 \cdot w^*(F)$: t=6.0, w*=True'}
CNF with n=10 violated $t(F) \leq 5 \cdot w^*(F)$: t=6.0, w*=True'}
TRIAL: {'metric_name': 'temperature_to_width_ratio', 'metric_value': 0, 'instances_tested': 1, 'conjecture_holds': False, 'counterexample': 'Random 3-CNF with n=10 violated $t(F) \leq 5 \cdot w^*(F)$: t=6.0, w*=True'}
CNF with n=10 violated $t(F) \leq 5 \cdot w^*(F)$: t=6.0, w*=True'}
TRIAL: {'metric_name': 'temperature_to_width_ratio', 'metric_value': 0, 'instances_tested': 1, 'conjecture_holds': False, 'counterexample': 'Random 3-CNF with n=8 violated $t(F) \leq 5 \cdot w^*(F)$: t=5.5, w*=True'}
CNF with n=8 violated $t(F) \leq 5 \cdot w^*(F)$: t=5.5, w*=True'}
TRIAL: {'metric_name': 'temperature_to_width_ratio', 'metric_value': 0, 'instances_tested': 1, 'conjecture_holds': False, 'counterexample': 'Random 3-CNF with n=8 violated $t(F) \leq 5 \cdot w^*(F)$: t=6.0, w*=True'}
CNF with n=8 violated $t(F) \leq 5 \cdot w^*(F)$: t=6.0, w*=True'}
TRIAL: {'metric_name': 'temperature_to_width_ratio', 'metric_value': 0, 'instances_tested': 1, 'conjecture_holds': False, 'counterexample': 'Random 3-CNF with n=8 violated $t(F) \leq 5 \cdot w^*(F)$: t=5.5, w*=True'}
CNF with n=8 violated $t(F) \leq 5 \cdot w^*(F)$: t=5.5, w*=True'}
TRIAL: {'metric_name': 'temperature_to_width_ratio', 'metric_value': 0, 'instances_tested': 1, 'conjecture_holds': False, 'counterexample': 'Random 3-CNF with n=8 violated $t(F) \leq 5 \cdot w^*(F)$: t=5.5, w*=True'}
CNF with n=8 violated $t(F) \leq 5 \cdot w^*(F)$: t=5.5, w*=True'}
```

## Judge reasoning

The ‘w’ field is reported as the boolean True rather than an integer tree-Resolution width, so the comparison  $t(F) \leq 5 \cdot w(F)$  is being evaluated against a coerced value of 1 rather than the actual refutation width. The reported falsifications are artifacts of a broken width oracle, not genuine counterexamples. | next: Fix the tree-Resolution width computation to return an integer (exact tree-DPLL depth/width), re-run the 30-seed protocol, and additionally verify the Tseitin lower-bound clause se

---

## Hochster Regularity of Variable Co-Occurrence Graph Lower-Bounds Resolution Width

- **Verdict:** INCONCLUSIVE
- **Bridge:** Castelnuovo-Mumford regularity of squarefree monomial edge ideals via Hochster’s combinatorial formula on independence complexes (Hochster 1977; Bruns-Herzog 1993; Katzman 2006; Woodroffe 2014) — a corner of computational commutative algebra with essentially no proof-complexity-side publications (arXiv search ‘edge ideal regularity’ AND (‘resolution width’ OR ‘proof complexity’) returns no hits)  $\times$  Resolution refutation width  $w^*(F)$  for unsatisfiable 3-CNFs, viewed as a Ben-Sasson-Wigderson and Krajíček-style stepping stone toward size lower bounds for tree-like Resolution and Extended Frege (via the depth- $O(1)$  Frege simulation of Resolution)



- **Recorded:** 2026-05-17 06:44 UTC
- **Entry ID:** e0a06bc1af46

## Statement

For an unsatisfiable 3-CNF  $F$  on  $n \leq 10$  variables, build the variable co-occurrence graph  $G(F)$  with  $V = \{x_1, \dots, x_n\}$  and edges  $\{x_i, x_j\}$  whenever  $x_i$  and  $x_j$  co-appear in at least one clause of  $F$ . Define  $\text{reg}(F) := \max\{j \geq 0 : \exists W \subseteq V \text{ with reduced rational homology } \tilde{H}\{|W|-j-1\}(\Delta\{G(F)[W]\}; \mathbb{Q}) \neq 0\}$ , where  $\Delta\{G[W]\}$  is the independence complex of the induced subgraph  $G[W]$  (Hochster's formula for  $\text{reg}(\mathbb{Q}[x]/I_{\{G(F)\}})$ ). Conjecture: for every unsatisfiable 3-CNF  $F$ , the minimum Resolution refutation width satisfies  $w(F) \geq \text{reg}(F) + 2$ ; *equivalently, a single  $F$  with  $w(F) < \text{reg}(F) + 2$  falsifies it.*

## Rationale

Hochster’s formula recasts an algebraic invariant of the edge ideal as the maximum homological complexity over induced subgraphs, identifying variable clusters whose mutual logical entanglement is topologically nontrivial; intuitively, no narrow refutation can simultaneously break such a cluster, forcing high resolution width. The bridge from edge-ideal regularity to proof complexity is essentially unstudied, and unlike single-number truth-table invariants (calibration trap (j)),  $\text{reg}(F)$  is a *relational* invariant of the CNF graph computed by quantifying over induced substructures, sidestepping the Razborov–Rudich largeness criterion that defeats truth-table approaches. If valid, BSW gives size  $\geq \exp(\Omega(\text{reg}(F)^2/n))$ , and via Krajíček’s depth-d Frege simulation this propagates to nontrivial EF line-count certificates.

## Novelty

- Judge: NOVEL over 0 arXiv hits

## Empirical Test

- exit code: 1, elapsed: 0.15s

```
TRIAL: {'metric name': 'w*(F) - reg(F)', 'metric value': 0, 'instances tested': 1, 'conjecture holds': True}
```

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0d752578.py", line 149, in <module>
 result = run_trial(seed)
            ~~~~~^~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0d752578.py", line 133, in run_trial
    w = resolution_width(F, n)
        ~~~~~^~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0d752578.py", line 108, in resolution_w
 if any((var, True) in c1 and (var, False) in c2) or any((var, False) in c1 and (var, True) in c2):
    ~~~~~
```

```
TypeError: 'bool' object is not iterable
```

## Judge reasoning

The test crashed with a `TypeError` in `resolution_width` (misuse of `any()` on a boolean expression), producing only 1 trial instead of the pre-registered 90 seeds. No valid data was collected to evaluate the support condition. | next: Fix the `resolution_width` function (the `any()` calls wrap non-iterable boolean expressions) and rerun the full 90-seed sweep across  $n \in \{6, 8, 10\}$ .

## Ihara Zeta Entropy Predicts Tseitin DPLL Tree Size

- **Verdict:** INCONCLUSIVE
- **Bridge:** Ihara zeta function of finite graphs (Hashimoto-Stark-Terras combinatorial graph-zeta theory rooted in number-theoretic L-functions; essentially no papers connect Ihara zeta values to proof complexity, distinct from non-backtracking spectral approaches) × Tree-like Resolution / DPLL refutation tree size for Tseitin XOR formulas  $T(G, \omega)$  on 3-regular graphs (Urquhart-Ben-Sasson-Razborov-Atserias regime)
- **Recorded:** 2026-05-17 07:23 UTC
- **Entry ID:** 3ba5471eff37

### Statement

Let  $G$  be a connected 3-regular graph on  $n$  vertices with adjacency matrix  $A$ , and let  $\omega: V \rightarrow \{0,1\}$  have  $\sum \omega$  odd, so  $T(G, \omega)$  is unsatisfiable. Define the Ihara entropy  $H(G) := (1/n) \cdot \ln \det( (11/9) \cdot I - A/3 )$ , which is well-defined and positive because  $(11/3) \cdot I - A$  has eigenvalues in  $[2/3, 20/3]$  for any 3-regular  $G$ . We conjecture that with pre-registered absolute constants  $c_1 = 1/20$  and  $c_2 = 20$ , every connected 3-regular  $G$  with odd  $\omega$  satisfies  $c_1 \cdot n \cdot H(G) - 1 \leq \log_2( S\_DPLL(T(G, \omega)) ) \leq c_2 \cdot n \cdot H(G) + \log_2(n) + 1$ , where  $S\_DPLL$  is the leaf count of the canonical DPLL refutation tree using unit propagation plus first-unassigned-variable branching. A single  $(G, \omega)$  violating either inequality refutes the conjecture.

### Rationale

The Ihara zeta function  $\zeta_G(u) = \prod_p (1 - u^{|p|})^{-1}$  over prime non-backtracking cycles is encoded by  $\det(I - uA + u^2(D - I))$ ; at  $u = 1/3$  this determinant aggregates how cycles of all lengths contribute multiplicatively, and each cycle of  $G$  corresponds in  $T(G, \omega)$  to a closed XOR constraint that DPLL must locally certify, so per-vertex cycle entropy should track per-vertex branching entropy. The evaluation point  $u = 1/3$  sits strictly inside the radius of convergence for 3-regular graphs, giving a numerically stable, integer-matrix invariant whose growth on Ramanujan-type expanders matches Urquhart's exponential refutation-size lower bound. This bridges number-theoretic graph zetas to proof complexity in a regime where, to our knowledge, fewer than five papers have looked.

### Novelty

- Judge: NOVEL over 0 arXiv hits

### Empirical Test

- exit code: 124, elapsed: 240.01s

TIMEOUT after 240s

### Judge reasoning

The test timed out after 240s (returncode=124) without producing a RESULT line or any instance data, so neither the bound inequalities nor the Spearman  $\rho$  threshold could be evaluated. | next: Reduce the instance budget (e.g. fewer seeds or cap  $n$  at 12) and/or replace the exact DPLL leaf count with an iterative bounded search so the 150-instance sweep completes within the timeout.

---

## Signed-Support $L_\infty$ Discrepancy Bounds Bottom Fan-In for AC0 PARITY

- **Verdict:** INCONCLUSIVE
- **Bridge:** Combinatorial discrepancy theory (Beck-Fiala, Spencer six-standard-deviations, Chazelle-Matousek hereditary discrepancy of  $\pm 1$  incidence matrices) ×  $AC^0$  depth- $d$  circuit size and bottom fan-in for PARITY $_n$ , in the Hastad / Yao switching-lemma regime

- **Recorded:** 2026-05-17 08:06 UTC
- **Entry ID:** 2f193a8b2209

### Statement

For an  $AC^0$  circuit  $C$  on  $n$  inputs with bottom-level AND/OR gates  $B_1, \dots, B_m$ , define the signed support matrix  $M(C)$  in  $\{-1, 0, +1\}^{m \times n}$  by  $M_{ij} = +1$  if  $x_j$  appears positively in  $B_i$ ,  $-1$  if negatively, and  $0$  if absent, and let  $w(C) = \max_i ||M_i||_0$  be the maximum bottom fan-in. Define the  $L_{\infty}$  discrepancy  $\psi(C) := \min\{\chi \in \{-1, +1\}^n \mid \max_{i \leq m} |\sum_j M_{ij} \chi_j|\}$ . CONJECTURE: there is an absolute constant  $c \geq 1/4$  such that for every  $AC^0$  circuit  $C$  of any constant depth that computes  $PARITY_n$ ,  $\psi(C) \geq c * w(C)$ ; a single  $PARITY$ -computing  $AC^0$  instance with  $\psi(C) < (1/4) * w(C)$  refutes it.

### Rationale

$PARITY$  forces every bottom-gate sign pattern to participate in a covering of  $(n-1)$ -parity-correct assignments, so the rows of  $M(C)$  must span a Hadamard-like sign system that admits no balanced coloring whose row-sums all fall below the typical  $\sqrt{w}$  scale; this is a structural cancellation obstruction distinct from random restrictions. The invariant is exactly  $S_n \times Z_2^n$  equivariant (variable relabeling and literal-polarity flips both preserve  $\psi$  and  $w$ ), satisfying the focus-mode naturality demand. If correct, the bound  $\psi \geq c*w$  replaces the switching-lemma probability calculation by a deterministic discrepancy certificate and is sharp on the canonical CNF/DNF (where  $\psi \sim n = w$ ).

### Novelty

- Judge: NOVEL over 0 arXiv hits

### Empirical Test

- exit code: 124, elapsed: 240.10s

TIMEOUT after 240s

### Judge reasoning

Test timed out after 240s (returncode 124) before producing any RESULT line or data; the pre-registered support condition cannot be evaluated. | next: Reduce search space (e.g., restrict to  $n$  in  $\{4, 6, 8\}$  and fewer seeds) or optimize  $\psi$  computation via branch-and-bound/SDP relaxation to obtain data within the time budget.

---

## 2-Sylow Rank of Critical Group Bounds Tseitin DPLL Size

- **Verdict:** INCONCLUSIVE
- **Bridge:** Algebraic combinatorics of the critical (sandpile) group  $K(G)$ : specifically the 2-Sylow rank  $r_2(G) = \dim_{\mathbb{F}_2} \ker(\tilde{L}_G \bmod 2)$  of the reduced graph Laplacian, in the Björner-Lovász-Shor / Wood-Mészáros tradition on cokernels of random integer matrices; distinct from the well-studied invariant  $|K(G)| = \tau(G)$  and essentially absent from Tseitin proof-complexity literature  $\times$  Tree-like Resolution / DPLL refutation size for Tseitin XOR formulas  $T(G, \omega)$  on 3-regular graphs (Urquhart-Ben-Sasson-Razborov regime), via the cycle-space-mod-2 mechanism underlying expander Tseitin lower bounds
- **Recorded:** 2026-05-17 08:50 UTC
- **Entry ID:** eb9e15c9c9db

## Statement

For every connected 3-regular graph  $G$  on  $n$  vertices and every charge  $\omega: V \rightarrow \mathbb{F}_2$  with  $\sum \omega \equiv 1 \pmod{2}$ , let  $\tilde{L}G$  be the reduced graph Laplacian (Laplacian with one row/column removed) and define the 2-Sylow rank  $r_2(G) := (n-1) - \text{rank}\{\tilde{L}G \pmod{2}\}$ . Let  $N_{\text{DPLL}}(T(G, \omega))$  be the number of nodes in the canonical DPLL refutation tree of the Tseitin CNF (with unit propagation, branch on lowest-index unassigned variable). Conjecture:  $\log_2 N_{\text{DPLL}}(T(G, \omega)) \geq (1/4) \cdot (n-1-r_2(G))$  for every such  $(G, \omega)$ ; equivalently, a single  $(G, \omega)$  with  $\log_2 N_{\text{DPLL}} < (1/4)(n-1-r_2(G))$  falsifies it.

## Rationale

Tseitin refutation hardness is governed by the cycle-space mod 2 of  $G$ , and  $r_2(G)$  precisely captures how many independent  $\mathbb{F}_2$ -degenerate directions the Laplacian pairing has on that cycle space; thus low  $r_2(G)$  (full  $\mathbb{F}_2$ -rigidity) should force DPLL to traverse all cycle-space cosets while high  $r_2$  opens algebraic short-cuts. This invariant is graph-structural (not a truth-table predicate), bypassing natural proofs, and is computed by elementary  $\mathbb{F}_2$  linear algebra rather than ring extensions, so it does not algebrize. It connects an under-utilised corner of algebraic combinatorics (2-part of Jacobian cokernel distributions) to the established expander-Tseitin lower-bound machinery.

## Novelty

- Judge: NOVEL over 0 arXiv hits

## Empirical Test

- exit code: 124, elapsed: 240.10s

TIMEOUT after 240s

## Judge reasoning

The test timed out after 240s (returncode 124) without producing a RESULT line or any instance data, so the pre-registered support condition over 150 instances cannot be evaluated. | next: Re-run with a longer timeout and/or reduce the size range (e.g.,  $n \in \{6, 8, 10, 12\}$ ) and cache DPLL counts to obtain complete data on all 150 instances.

---

## Stable-Rank of Layer-Symbol Matrix Separates Read-Twice BPs for IP<sub>2</sub>

- **Verdict:** INCONCLUSIVE
- **Bridge:** Forster-style stable-rank / numerical-rank theory (Forster 2002; Linial-Shraibman 2009) — a corner of sign-rank / discrepancy-based lower-bound machinery transplanted from communication matrices to the operator-valued symbol bundle of a branching program; this bridge has essentially no presence in BP literature outside the unrelated Nechiporuk rank tradition. × Read-twice oblivious branching program size for IP<sub>2</sub> on  $2n$  adversarial-ordered variables (Borodin-Razborov-Smolensky 1993 open separation), in the meta-complexity sense of seeking a ‘compressibility witness’  $\rho(P)$  with  $\rho(P) = O(\log \text{size}(P))$  — a poly-time-computable structural invariant of the BP, NOT of its truth-table.
- **Recorded:** 2026-05-17 09:02 UTC
- **Entry ID:** 46c3da6284c4

## Statement

For a read-twice oblivious BP  $P$  of size  $s$  on  $2n$  variables, with layers  $t=1..4n$  and transition matrices  $M_t(b) \in \{0,1\}^{s \times s}$  for  $b \in \{0,1\}$ , form the layer-symbol matrix  $S(P) := [\Delta_1 \mid \Delta_2 \mid \dots \mid \Delta_{4n}]$

$\in \mathbb{R}^{\{s \times 4n \cdot s\}}$  where  $\Delta_t := M_t(1) - M_t(0)$ . Define the stable-rank invariant  $\rho(P) := \|S(P)\|_{F^2} / \|S(P)\|_{op^2}$ . Conjecture: (i)  $\rho(P) \leq 2 \cdot s$  for every read-twice BP  $P$  (numerical-rank anchor: this follows from  $\text{rank} \leq s$  but we conjecture the tighter constant 2 holds for all syntactic BPs); (ii) for every read-twice BP  $P$  with output correlation  $\langle f_P, IP_2 \rangle \geq \varepsilon$  over uniform  $2n$ -bit inputs,  $\rho(P) \geq c \cdot \varepsilon^2 \cdot 2^{\{n/2\}}$  for an absolute constant  $c > 0$ . A single instance with  $\langle f_P, IP_2 \rangle \geq \varepsilon$  and  $\rho(P) < c \cdot \varepsilon^2 \cdot 2^{\{n/2\}}$ , or any read-twice BP with  $\rho(P) > 2s$ , refutes it.

## Rationale

Each  $\Delta_t$  records how flipping one read at layer  $t$  shifts the BP's state operator; for a small BP these shifts share an  $s$ -dimensional column space, capping the stable rank at  $O(s)$ .  $IP_2$ 's adversarial pairing of  $x_i$  with  $y_i$  forces the symbol bundle to encode  $2^{\{\Theta(n)\}}$  mutually-orthogonal cross-variable shift directions (after the middle cut, each pair  $(x_i, y_i)$  contributes a Hadamard-flavoured outer product), so the Frobenius mass spreads across exponentially many comparable singular values, making  $\|S\|_{F^2} / \|S\|_{op^2}$  grow geometrically with  $n$ . The setup avoids natural-proofs ( $\rho$  is a structural property of  $P$ , not its truth-table), algebrization (singular values are analytic, not ring-arithmetic), and relativization (no oracle).

## Novelty

- Judge: NOVEL over 6 arXiv hits

Top hits consulted: - [0710.5926v2] Mod 2 cohomology of 2-local finite groups of low rank - [1503.07648v2] Sign rank versus VC dimension - [1708.02054v1] Pseudorandom Bits for Oblivious Branching Programs - [0809.2093v2] An approximation algorithm for approximation rank - [1706.02207v3] On The Communication Complexity of High-Dimensional Permutations

## Empirical Test

- exit code: 0, elapsed: 0.08s

```
551269602595 > 2s=8", "seed": "421"}
TRIAL: {"metric_name": "stable_rank", "metric_value": 87.00000000000001, "metric_std": 0.0, "instances_tested": 0, "metric_max": 87.00000000000001, "metric_min": 87.00000000000001}
TRIAL: {"metric_name": "stable_rank", "metric_value": 80.24174837076876, "metric_std": 0.0, "instances_tested": 0, "metric_max": 80.24174837076876, "metric_min": 80.24174837076876}
TRIAL: {"metric_name": "stable_rank", "metric_value": 39.25023292246046, "metric_std": 0.0, "instances_tested": 0, "metric_max": 39.25023292246046, "metric_min": 39.25023292246046}
TRIAL: {"metric_name": "stable_rank", "metric_value": 47.12182530062137, "metric_std": 0.0, "instances_tested": 0, "metric_max": 47.12182530062137, "metric_min": 47.12182530062137}
TRIAL: {"metric_name": "stable_rank", "metric_value": 65.81197395850668, "metric_std": 0.0, "instances_tested": 0, "metric_max": 65.81197395850668, "metric_min": 65.81197395850668}
TRIAL: {"metric_name": "stable_rank", "metric_value": 0.0, "metric_std": 0.0, "instances_tested": 0, "metric_max": 0.0, "metric_min": 0.0}
TRIAL: {"metric_name": "stable_rank", "metric_value": 36.561209216717984, "metric_std": 0.0, "instances_tested": 0, "metric_max": 36.561209216717984, "metric_min": 36.561209216717984}
TRIAL: {"metric_name": "stable_rank", "metric_value": 28.50532667271179, "metric_std": 0.0, "instances_tested": 0, "metric_max": 28.50532667271179, "metric_min": 28.50532667271179}
TRIAL: {"metric_name": "stable_rank", "metric_value": 59.148813421486146, "metric_std": 0.0, "instances_tested": 0, "metric_max": 59.148813421486146, "metric_min": 59.148813421486146}
```

## Judge reasoning

The critic correctly notes that  $\rho(P) = \|S\|_{F^2} / \|S\|_{op^2}$  is mathematically bounded by  $\text{rank}(S) \leq s$ , so reported values like 26–129 with  $2s=8$  are impossible and indicate a metric-implementation bug (likely summing per-layer ratios instead of computing the stable rank of the concatenated matrix). The pre-registered support condition is therefore not validly evaluated, and the ‘falsification’ cannot be trusted. | next: Fix the  $\rho$  computation to actually compute  $\|S\|_{F^2} / \|S\|_{op^2}$  on the concatenated  $S \in \mathbb{R}^{\{s \times 4n\}}$

## TP\_2 Minor-Sign Defect of Spectral Pseudo-Moment Matrix Bounds Max-CUT Ratio

- Verdict: INCONCLUSIVE

- **Bridge:** Karlin-Edrei-Schoenberg total positivity / Pólya frequency theory (TP<sub>k</sub> matrices, sign-regularity of 2x2 minors) × Degree-2 SOS pseudo-moment matrix for Max-CUT on graphs (Goemans-Williamson regime), with a lifting interpretation via gadget composition
- **Recorded:** 2026-05-17 09:29 UTC
- **Entry ID:** 16809e692923

## Statement

For a connected graph  $G=(V,E)$  on  $n$  vertices with combinatorial Laplacian  $L=D-A$ , let  $v_2, v_3 \in \mathbb{R}^n$  be the second and third eigenvectors of  $L$  (unit-norm, breaking ties by lexicographic sign). Define the rank-2 spectral pseudo-moment matrix  $M_G[i,j] := (v_2[i]v_2[j] + v_3[i]v_3[j])/Z$ , where  $Z := \max\{a,b\}|v_2[a]v_2[b] + v_3[a]v_3[b]|$ , and the TP<sub>2</sub> defect  $\delta(G) := |\{(i < j, k < l) : \det M_G[\{i,k\}, \{j,l\}] < -1/n^3\}| / \binom{n,2}^2$ . Conjecture: there exist absolute constants  $c_1=0.5$  and  $c_2=0.10$  such that (i) every graph with  $\max\text{-CUT}(G)/|E| \geq 0.879$  satisfies  $\delta(G) \leq c_1 \cdot n^{-1/2}$ , and (ii) every graph on  $n \geq 12$  vertices with  $\max\text{-CUT}(G)/|E| \leq 0.55$  satisfies  $\delta(G) \geq c_2$ . A single graph violating either inequality refutes the conjecture.

## Rationale

Pólya frequency / TP<sub>2</sub> matrices are exactly the moment matrices of log-concave one-parameter families; SOS-tight max-CUT instances (where degree-2 SOS already certifies a near-optimal cut) should correspond to log-concave 1D rounding directions and thus preserve TP<sub>2</sub> in the leading spectral block, whereas integrality-gap instances spread mass in a way that flips 2x2 minor signs. Because TP<sub>2</sub> is closed under coordinatewise polynomial gadget tensoring (Karlin 1968, Ch. 3),  $\delta$  behaves as a lifting-friendly invariant: gadget composition can only grow  $\delta$  by a constant factor, matching the ‘preserved under polynomial reductions’ requirement on the target property P. The bridge between Pólya frequency theory and SOS Max-CUT lower bounds appears to have under five papers (TP<sub>k</sub> has been used in stable-polynomial proofs of Kadison-Singer but not in SOS degree-lower-bound machinery).

## Novelty

- Judge: NOVEL over 0 arXiv hits

## Empirical Test

- exit code: 1, elapsed: 3.25s

TRIAL: {'metric\_name': 'TP\_2 defect', 'metric\_value': 0.0004911287328869746, 'instances\_tested': 8,

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_41095b21.py", line 253, in <module>
    result = run_trial(seed)
            ^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_41095b21.py", line 216, in run_trial
    delta = tp2_defect(graph)
            ^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_41095b21.py", line 186, in tp2_defect
    M = spectral_pseudo_moment_matrix(graph)
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_41095b21.py", line 174, in spectral_pse
    eigenvalues, eigenvectors = matrix_eigen(L)
                                ^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_41095b21.py", line 80, in matrix_eigen
    Q, R = qr_decomposition(A)
            ^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_41095b21.py", line 96, in qr_decomposit
    Q[i][k] = A[i][k] / R[k][k]
               ~~~~~^~~~~~
```

ZeroDivisionError: float division by zero

## Judge reasoning

The test crashed with a ZeroDivisionError in the custom QR decomposition after only 8 instances, well short of the pre-registered ~600 instance protocol. Although a putative low-cut counterexample was reported, the run terminated abnormally and the support condition cannot be evaluated. | next: Replace the hand-rolled QR with numpy.linalg.eigh (or guard against zero pivots) and rerun the full  $4 \times 30 \times 5$  protocol before judging the conjecture.

---

## Gromov 4-Point $\delta$ of XOR-Lifted Row Metric Bounds DT Depth

- **Verdict:** INCONCLUSIVE
- **Bridge:** Coarse geometry / Gromov 4-point  $\delta$ -hyperbolicity of finite metric spaces (Gromov 1987; Bridson-Haefliger 1999; Coornaert 2014) — a relational metric invariant from geometric group theory with essentially no presence in communication complexity (search ‘Gromov hyperbolicity’ AND ‘communication complexity’ returns no direct hits; <5 papers in adjacent areas)  $\times$  Deterministic two-party communication complexity of XOR-gadget-lifted Boolean functions, in the Raz-McKenzie / Göös-Pitassi-Watson / Hatami-Hosseini-Pitassi query-to-communication lifting framework, accessed through the bipartite row metric of the lifted communication matrix  $M_{\{f \circ \text{XOR}_2\}}$
- **Recorded:** 2026-05-17 10:01 UTC
- **Entry ID:** 5ee62d63ba3c

## Statement

Let  $f: \{0,1\}^n \rightarrow \{0,1\}$  have deterministic decision-tree depth  $d = \text{DT}(f)$ , let  $g = \text{XOR}_2$  (Alice has  $x_i \in \{0,1\}$ , Bob has  $y_i \in \{0,1\}$ ,  $g(x_i, y_i) = x_i \oplus y_i$ ), and let  $M \in \{0,1\}^{\ell^2 n \times 2^n}$  be the lifted matrix  $M[x,y] = f(x \oplus y)$ . View rows of  $M$  as points in  $(\{0,1\}^{\ell^2 n}, d_H)$ . For each 4-tuple of distinct rows  $(a,b,c,d)$  compute the three sums  $S_1 = d_H(a,b) + d_H(c,d)$ ,  $S_2 = d_H(a,c) + d_H(b,d)$ ,  $S_3 = d_H(a,d) + d_H(b,c)$ , sort them  $S_{(1)} \geq S_{(2)} \geq S_{(3)}$ , and set  $\delta(M) = \max \text{ over 4-tuples of } (S_{(1)} - S_{(2)})/2$ ; let  $\delta(M) = \delta(M)/\text{diam}(M)$ . Conjecture: for every non-constant  $f$ ,  $\delta(M_{\{f \circ \text{XOR}_2\}}) \leq 1 - \text{DT}(f)/(n+1)$ , i.e. higher DT-depth forces strictly more 4-point-treelike row geometry; a single  $f$  with  $\delta(M) > 1 - \text{DT}(f)/(n+1)$  refutes it.

## Rationale

Lifting theorems claim DT structure of  $f$  survives into the gadget-amplified bipartite matrix; if so, the inherited combinatorial structure should constrain coarse-geometric quantities of the row Hamming cloud, since high-DT functions composed with  $\text{XOR}_2$  build hierarchical query trees that the Gromov 4-point inequality is precisely designed to detect. Crucially  $\delta$  is a 4-point relational invariant — not a single-number truth-table functional — so it sidesteps the Razborov-Rudich largeness obstruction, and uses only Hamming distances (no ring/polynomial structure), so it should not algebrize. If the bound holds, it gives a coarse-geometric proxy for the lifted communication lower bound via embedding-distortion arguments à la Linial-London-Rabinovich.

## Novelty

- Judge: NOVEL over 0 arXiv hits

## Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f5a45145.py", line 97, in <module>
 result = run_trial(seed)
            ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f5a45145.py", line 69, in run_trial
    s3 = distances[sum(range(a+1, d+1))] + distances[sum(range(b+1, c+1))]
            ~~~~~^~~~~~
```

IndexError: list index out of range

## Judge reasoning

The test crashed with an IndexError before producing any data, so neither the support condition nor a counterexample was established. Per the rules, a crash yields INCONCLUSIVE. | next: Fix the indexing bug in the 4-tuple distance lookup (use a proper pair-to-index mapping like distances[(i,j)] or itertools.combinations indexing) and rerun the pre-registered sweep.

---

## Median Half-Subgraph Separator Lower-Bounds Tseitin DPLL Size

- **Verdict:** INCONCLUSIVE
- **Bridge:** Separation profile of geometric group theory (Benjamini-Schramm-Timár 2012; Hume-Mackay-Tessera 2020; Hume 2017): the coarse-geometric invariant  $\text{sep}_G(k) = \sup$  over  $k$ -vertex subgraphs  $H$  of min balanced vertex separator of  $H$ , used to distinguish quasi-isometry classes of finitely generated groups. This corner of GGT has essentially no presence in proof complexity — an arXiv search ‘separation profile’ AND (‘resolution’ OR ‘Tseitin’ OR ‘proof complexity’) returns no hits; the closest neighbours are asymptotic-dimension papers that never reach proof systems. × Tree-like Resolution / DPLL refutation tree size for Tseitin XOR formulas  $T(G, \omega)$  on 3-regular graphs, in the Urquhart-Ben-Sasson-Wigderson-Krajíček bounded-arithmetic regime where  $\neg T(G, \omega)$  is a  $\Sigma^b_0$  statement and DPLL size lower-bounds  $V^0$ -witnessing complexity.
- **Recorded:** 2026-05-17 10:32 UTC
- **Entry ID:** 118fd4951a23

## Statement

For a connected 3-regular graph  $G$  on  $m$  vertices and odd charge  $\omega$ :  $\nu_{30}(G) := \text{median over } 30 \text{ uniformly random vertex subsets } U \subset V(G) \text{ of size } k=\lfloor m/2 \rfloor \text{ of the value } \beta(G[U])$ , where  $\beta(H)$  is the size of a minimum vertex separator  $(A, V(H) \setminus (A \cup S), S)$  of  $G[U]$  with  $|A|, |V(H) \setminus (A \cup S)| \geq \lfloor |V(H)|/3 \rfloor$ , and  $\beta(H) := 0$  if  $G[U]$  is disconnected. Conjecture: (i) every odd-charge Tseitin formula  $T(G, \omega)$  requires DPLL tree size  $s(T(G, \omega)) \geq 2^{\{0.1 \cdot \nu_{30}(G)\}}$ ; (ii) for every 3-regular  $G$  admitting a vertex partition  $(V_1, V_2)$  with  $|V_1|, |V_2| \geq m/3$  and  $|E(V_1, V_2)| \leq 2$  (e.g., barbells, prism-pairs joined by a bridge or two edges),  $\nu_{30}(G) \leq 2$ . A single 3-regular instance with  $\log_2(s) < 0.1 \cdot \nu_{30}(G)$ , or a small-cut graph in class (ii) with  $\nu_{30}(G) > 2$ , refutes the conjecture.

## Rationale

The Ben-Sasson-Wigderson width method bounds Tseitin Resolution length via expansion of  $G$  as a whole; but bounded arithmetic  $V^0$ -witnessing of unsatisfiable  $\Sigma^b_0$ -instances should react to every scale of expansion, not just the global one. The separation profile evaluated at half-size captures exactly this: it is large iff typical half-subgraphs of  $G$  are themselves robustly connected, which is the structural data DPLL must repeatedly re-encounter on any branch that fixes half the



variables. The link to Cook-Reckhow-Krajíček:  $\nu_{30}$  is a low-complexity (NC-computable up to sampling)  $\Sigma^b_0$ -definable property of  $G$ , so a separation ' $\nu_{30}$  large  $\Rightarrow$  refutation long' is exactly the bounded-arithmetic style of structural lower bound that escapes both NATURAL\_PROOFS ( $\nu_{30}$  depends on  $G$ , not on a truth-table) and ALGEBRIZATION (no ring extension respects vertex-separator counts).

## Novelty

- Judge: NOVEL over 0 arXiv hits

## Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_41e6d60a.py", line 194, in <module>
 result = run_trial(seed)
            ~~~~~^~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_41e6d60a.py", line 155, in run_trial
    nu_30 = compute_nu_30(edges, n)
            ~~~~~^~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_41e6d60a.py", line 68, in compute_nu_30
 nu_values.append(compute_min_vertex_separator(subgraph_edges, k))
                    ~~~~~^~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_41e6d60a.py", line 46, in compute_min_v
    if not is_connected(edges, n):
           ~~~~~^~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_41e6d60a.py", line 35, in is_connected
 graph[v].add(u)
    ~~~~~^~
```

KeyError: 7

## Judge reasoning

The test crashed with a KeyError in `is_connected` before producing any RESULT line, so neither support nor falsification conditions can be evaluated. The critic also challenged due to lack of data. | next: Fix the graph construction in `is_connected` (initialize adjacency entries for all vertices in the induced subgraph, e.g., using `defaultdict(set)` or pre-populating keys) and rerun the pre-registered protocol.

---

## Cubic-Form Lie-Stabilizer Codim Lower-Bounds DISJ Communication

- **Verdict:** BARRIER\_HIT
- **Bridge:** Geometric Complexity Theory: Mulmuley-Sohoni Lie-algebra stabilizers of explicit cubic forms  $c_M \in \text{Sym}^3(\mathbb{R}^N)$ , accessed through the codimension  $\kappa(M) = N^2 - \dim \text{Stab}_{\{gl_N\}}(c_M)$ . This is a first-order/infinitesimal GCT invariant distinct from (a) Kronecker-coefficient counts and (b) full orbit-closure dimensions, the two GCT-COMM bridges that previously hit ALGEBRIZATION in this approach; <5 papers connect cubic-form stabilizers to communication complexity.  $\times$  Randomized two-party communication complexity  $CC_R(M)$  of Boolean matrices  $M \in \{0,1\}^{\{N \times N\}}$ , with the target  $M = M_{\text{DISJ}_n}$  on  $N = 2^n$  inputs (Razborov 1992  $\Omega(n)$  bound), instantiating the COMM\_DISJ tensor-invariant program.
- **Recorded:** 2026-05-17 10:38 UTC
- **Entry ID:** 9d129ecb4a7f

## Statement

For  $M \in \{0,1\}^{\{N \times N\}}$  ( $N = 2^n$ ), form the symmetric cubic form  $c_M(x) := \sum_{i,j,k \in [N]} M[i,j] \cdot M[j,k] \cdot M[i,k] \cdot x_i x_j x_k$  whose coefficient at the symmetrized monomial  $x_i x_j x_k$  is the count of directed triangles  $(i,j,k)$  in the bipartite-merged graph of  $M$ . Let  $g(M) := \dim\{A \in \mathfrak{gl}_N : \text{the polynomial } \sum_i (Ax)_i \cdot \partial c_M / \partial x_i \text{ is identically zero}\}$ , and define the GCT stabilizer codimension  $\kappa(M) := N^2 - g(M)$ . Conjecture: (a) for every Boolean  $M$ ,  $CC\_R(M) \geq (1/8) \cdot \log_2(1 + \kappa(M))$ , refuted by any single matrix with a randomized log-rank-style upper bound  $CC\_R(M) \leq 2 \cdot \log_2(\text{rank}_R(M)) + 1$  violating  $(1/8) \cdot \log_2(1 + \kappa(M))$  by more than 1; (b)  $\kappa(M\_DISJ\_n) \geq N - N^{\{3/4\}}$  for all  $n \geq 3$ , so (a) certifies  $CC\_R(DISJ\_n) \geq n/8 - O(1) = \Omega(n)$ .

## Rationale

The triangle-count cubic form  $c_M$  encodes a non-polynomial-extension-friendly combinatorial feature of  $M$  (counting 3-cycles in its associated digraph) that is invisible to algebraic oracles, addressing why prior Kronecker/orbit-dimension GCT-COMM attempts hit ALGEBRIZATION. The infinitesimal Lie stabilizer is a first-order Mulmuley-Sohoni invariant that, unlike rank, is sensitive to triangle-richness — and DISJ’s communication matrix has a sparse 3-disjoint structure ( $|\{(S,T,U): \text{pairwise disjoint}\}| = 3^n \square N^3$ ) that should generically force a trivial stabilizer. The  $1/8$  constant tracks a sym-cubic Hilbert-series degeneracy factor and is conservative.

## Novelty

- Judge: “ over 0 arXiv hits

## Empirical Test

- exit code: -1, elapsed: 0.00s

(no output)

## Judge reasoning

Rejected by ALGEBRIZATION barrier

---

## Discrete Morse Critical Cells of Conflict Complex Bound DPLL Tree Size

- **Verdict:** INCONCLUSIVE
- **Bridge:** Discrete Morse theory on simplicial complexes (Forman 1998; Joswig-Pfetsch 2005 greedy lex matching; Benedetti-Lutz 2014 random discrete Morse) — a corner of combinatorial topology essentially absent from proof complexity (arXiv ‘discrete Morse’ AND (‘Resolution’ OR ‘DPLL’ OR ‘proof complexity’) returns no direct hits, with <5 adjacent independence-complex papers)  $\times$  DPLL / tree-like Resolution refutation tree size  $t(F)$  for unsatisfiable 3-CNFs, in the Cook-Reckhow-Krajíček bounded-arithmetic regime where  $\neg F$  is a  $\Sigma^b_0$  statement and  $\log_2 t(F)$  controls  $V^0$ -witnessing complexity of unsatisfiability
- **Recorded:** 2026-05-17 11:47 UTC
- **Entry ID:** 02fea1709cc2

## Statement

For an unsat 3-CNF  $F$  with clauses  $c_1, \dots, c_m$ , let  $C_F$  be the clause-conflict graph (vertex per clause; edge  $\{c_i, c_j\}$  iff they share at least one complementary literal pair) and let  $I_{\leq 3}(C_F)$  be the independence complex of  $C_F$  truncated to faces of size  $\leq 4$  (faces = clause-subsets pairwise free of complementary-literal conflicts). Run Joswig-Pfetsch greedy lex discrete-Morse matching: traverse simplices in (dim, lex) order, pair each  $\sigma$  with its lex-smallest currently-unmatched coface

$\tau$  of  $\dim(\sigma)+1$  inside the truncation (otherwise mark  $\sigma$  critical). Let  $\delta(F) :=$  number of critical cells of dimension  $\geq 1$ . Conjecture: there is an absolute constant  $c \geq 1/4$  such that for every unsat 3-CNF  $F$  with  $m \geq 4$ ,  $\log_2(t(F)) \geq c \cdot \log_2(\delta(F)+2)$ , where  $t(F)$  is the DPLL refutation tree-leaf count under lex-variable branching with unit propagation. A single  $F$  with measured  $\log_2(t^*(F)) < 0.25 \cdot \log_2(\delta(F)+2)$  refutes it.

## Rationale

Critical cells of  $I(C_F)$  count obstructions to extending a locally-consistent partial certificate (a set of mutually non-resolving clauses) to a global one, which is precisely the local-to-global obstruction Ben-Sasson-Wigderson width-size trade-offs quantify for DPLL. A logarithmic relation is the natural scale:  $t^*$  grows exponentially in hardness while  $\delta$  grows exponentially in conflict-topological richness, so log-log alignment is plausible. The bridge avoids algebrization (Forman matching is purely combinatorial — no ring, no polynomial extension), evades natural proofs (the invariant is syntactic on  $F$ , not on its truth table), and lies in a regime Razborov/Krajíček bounded-arithmetic witnessing leaves open.

## Novelty

- Judge: NOVEL over 0 arXiv hits

## Empirical Test

- exit code: 0, elapsed: 0.04s

7}

```

TRIAL: {'metric_name': 'log2(t*)/log2(δ+2)', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'log2(t*)/log2(δ+2)', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture
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TRIAL: {'metric_name': 'log2(t*)/log2(δ+2)', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture
RESULT: INCONCLUSIVE reason=no_valid_instances

```

## Judge reasoning

parse\_fail | next: NONE

## Gate-Cone Poset Mobius Value Lower-Bounds ACC<sup>0</sup> Size for MOD-q

- **Verdict:** INCONCLUSIVE
- **Bridge:** Mobius functions on incidence algebras of finite posets (Rota 1964 combinatorial inversion; Stanley EC1 Ch.3) applied to the gate-dependency poset of an ACC<sup>0</sup> circuit; an algebraic-combinatorial invariant essentially absent from circuit complexity (arXiv search ‘Mobius function poset’ AND ‘ACC<sup>0</sup>’ returns no hits; <5 adjacent papers on poset invariants for circuits)  $\times$  ACC<sup>0</sup>[m] circuit size for explicit symmetric functions (notably MOD<sub>q</sub> for  $q$  coprime to  $m$ ) in the Williams 2011 / Murray-Williams 2018 algorithmic-method regime, where structural circuit invariants underlie speedup-style lower bounds ( $\text{NEXP} \not\subseteq \text{ACC}^0$ )

- **Recorded:** 2026-05-17 15:23 UTC
- **Entry ID:** 3b280d479739

## Statement

Let  $C$  be an  $\text{ACC}^0[m]$  circuit on  $n$  inputs with  $s$  gates and depth  $d$ . Adjoin a formal minimum  $0$  below all gates and let  $P(C)$  be the poset on  $\{0\} \cup \text{Gates}(C)$  with  $g \leq h$  iff  $h = g$  or there is a directed wire path  $g \rightarrow h$  (and  $0 < g$  for every gate  $g$ ). Define  $M(C) := |\mu_{P(C)}(0, g_{\text{out}})|$  where  $\mu$  is the Mobius function of  $P(C)$  and  $g_{\text{out}}$  is the output gate, computed by  $\mu(0,0)=1$  and  $\mu(0,h) = -\sum_{0 \leq z < h} \mu(0,z)$ . Conjecture (A, upper anchor):  $M(C) \leq (s+1)^d$  for every  $\text{ACC}^0[m]$  circuit. Conjecture (B, hardness anchor): for every  $\text{ACC}^0[m]$  circuit  $C$  that correctly computes  $\text{MOD}_q$  on  $n$  inputs with  $\gcd(q,m)=1$ , one has  $M(C) \geq 2^{\{c \cdot n\} \{1/(d+1)\}}$  for an absolute constant  $c \geq 1/8$ ; a single counterexample circuit refutes (B).

## Rationale

Mobius inversion on the gate-dependency poset is a CIRCUIT invariant (not a truth-table property), so it dodges the natural-proofs trap that killed the Plunnecke variant in this approach; it is also non-ring (no polynomial extension makes the recursion  $\mu(0,h) = -\sum \mu(0,z)$  meaningful over  $F_p[X]$ ), so it should not algebrize. Heuristically, Smolensky's low-degree-polynomial obstruction for  $\text{MOD}_q$  in  $\text{ACC}^0[m]$  forces long, non-factorisable dependency chains in any small circuit, which inflate alternating-sign Mobius cancellations on antichains of the gate poset; the  $M(C)$  invariant captures this 'inversion depth' more sharply than gate count alone and is computable in  $O(s^2)$  time, making it a viable Williams-style structural witness.

## Novelty

- Judge: NOVEL over 0 arXiv hits

## Empirical Test

- exit code: 1, elapsed: 0.10s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_fbcfc841.py", line 123, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_fbcfc841.py", line 78, in run_trial
    circuit = generate_random_circuit(n, s, d, seed)
             ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_fbcfc841.py", line 27, in generate_random_circuit
    num_inputs = random.randint(2, min(3, len(gates)))
                ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/usr/lib/python3.12/random.py", line 336, in randint
    return self.randrange(a, b+1)
           ^^^^^^^^^^^^^^^^^^^^^
```

```
File "/usr/lib/python3.12/random.py", line 319, in randrange
    raise ValueError(f"empty range in randrange({start}, {stop})")
ValueError: empty range in randrange(2, 1)
```

## Judge reasoning

The test crashed with a ValueError in generate\_random\_circuit (empty range in randrange(2,1)) before any instances were evaluated, so neither the support nor falsification condition could be assessed. | next: Fix the circuit generator to guarantee  $\text{len}(\text{gates}) \geq 2$  (or clamp the fan-in lower bound) before sampling num\_inputs, then rerun the pre-registered sweep.

---

## F<sub>2</sub>-Corank of Minterm Incidence Bounds Monotone Formula Size

- **Verdict:** INCONCLUSIVE
- **Bridge:** Matroid theory — F<sub>2</sub>-linear matroid corank (dimension of left F<sub>2</sub>-null space) of the minterm-variable incidence matrix of a monotone Boolean function, viewed as a functional combinatorial-matroid invariant in the Oxley / Brualdi tradition of binary-matroid representability. This bridge to monotone formula complexity has essentially no presence in the literature (arXiv ‘minterm incidence F<sub>2</sub> corank’ AND ‘monotone formula’ returns no direct hits, <5 adjacent Razborov-approximator papers). × Monotone formula leaf-size L(f) for the k-CLIQUE indicator on K<sub>v</sub> (n = v(v-1)/2 edge variables) in the Razborov 1985 regime, with k = 3 fixed and the  $n^{\Omega(k)} \{1/4\}$  monotone lower bound as the prototypical target for the MONOTONE\_CLIQUE focus.
- **Recorded:** 2026-05-17 15:55 UTC
- **Entry ID:** dfc077cb4c36

### Statement

For a monotone Boolean function  $f$  on  $n$  variables, let  $M(f) \subset \{0,1\}^n$  be its set of minterms (each minterm identified with the 0/1 incidence vector of its support set), let  $A(f) \in F_2^{\{|M(f)| \times n\}}$  be the minterm-variable incidence matrix, and define  $\mu_2(f) := |M(f)| - \text{rank}_{\{F_2\}}(A(f))$  (the F<sub>2</sub>-matroid corank of the row matrix; equivalently the dimension of the left F<sub>2</sub>-null space). We conjecture (a)  $\mu_2(f) \leq L(f) \cdot \log_2(L(f)+2)$  for every monotone Boolean function  $f$ , where  $L(f)$  is the minimum monotone formula leaf-size of  $f$ ; and (b) for the  $k=3$  CLIQUE indicator on  $K_v$ ,  $\mu_2(f) = C(v,3) - (v-1)(v-2)/2 = (v-1)(v-2)(v-3)/6 = \Omega(v^3) = \Omega(n^{\{3/2\}})$ . Combined, (a)+(b) imply  $L(\text{CLIQUE}_{\{v,3\}}) = \Omega(v^3 / \log v)$ ; a single monotone  $f$  with  $\mu_2(f) > L(f) \cdot \log_2(L(f)+2)$  refutes (a), and any  $v \in \{4, \dots, 8\}$  with  $\mu_2(\text{CLIQUE}_{\{v,3\}}) \neq (v-1)(v-2)(v-3)/6$  refutes (b).

### Rationale

F<sub>2</sub>-corank of the minterm-variable incidence is a purely combinatorial matroid invariant that quantifies forced linear redundancies in the minterm cloud — exactly the kind of Steiner-design structure that the triangle hypergraph of  $K_v$  exhibits because each edge sits in  $v-2$  triangles. Razborov’s approximator philosophy already hints that bottom-up monotone formula construction cannot manufacture such global multiplicative redundancy without paying in leaves; turning that intuition into a clean matroid-rank bound (a) would yield a poly-time-computable, monotone-formula-tight, Steiner-aware lower-bound measure that contributes directly to the MONOTONE\_CLIQUE target while staying inside the algebraic-rank toolkit that previously yielded the SUPPORTED ‘Tropical Rank of Clause-Indicator’ result.

### Novelty

- Judge: NOVEL over 0 arXiv hits

### Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ca306781.py", line 100, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ca306781.py", line 88, in run_trial
    "metric_value": sum(r["mu_2"] for r in results),
                    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ca306781.py", line 88, in <genexpr>
    "metric_value": sum(r["mu_2"] for r in results),
                      ~^^^^^^^^
```

KeyError: 'mu\_2'

## Judge reasoning

The test crashed with a KeyError: 'mu\_2' before producing any data, so no RESULT line was emitted and neither the support nor falsification condition can be evaluated. | next: Fix the test harness so each result dict includes the 'mu\_2' key (or guard the sum with .get), then re-run the pre-registered protocol over  $v \in \{4..8\}$  and the random monotone/TH\_k^n configs.

---

## Hypercontractive Laplacian-Spectrum Flatness Bounds Spectral SOS Max-Cut Ratio

- **Verdict:** INCONCLUSIVE
- **Bridge:** Fourier-analytic hypercontractive flatness invariants of graph Laplacian spectra (4-norm-to-2-norm ratios in the Bonami-Beckner-Bourgain tradition, applied as a 'pseudo-Fourier-level-2' invariant of the degree-2 SOS moment-matrix proxy)  $\times$  Spectral / degree-2 SOS moment-matrix relaxation of Max-Cut on random 3-regular graphs (Delorme-Poljak spectral upper bound  $(n/4)\lambda_{\max}$  as the canonical degree-2 SOS surrogate; integrality gap target  $\geq 0.121$ )
- **Recorded:** 2026-05-17 16:22 UTC
- **Entry ID:** c8a7a640d375

## Statement

For a 3-regular graph  $G$  on  $n$  vertices with Laplacian  $L = D - A$  having eigenvalues  $0 = \lambda_n < \lambda_{n-1} \leq \dots \leq \lambda_1$ , define the hypercontractive Laplacian flatness  $\text{HCF}(G) := (n-1) \cdot (\sum_{k=1}^{n-1} \lambda_k^4) / (\sum_{k=1}^{n-1} \lambda_k^2)^2 \in [1, n-1]$ , and the spectral degree-2 moment-matrix Max-Cut ratio  $\rho(G) := \text{MC}(G) / \text{SB}(G)$ , where  $\text{SB}(G) := (n/4) \cdot \lambda_1$  and  $\text{MC}(G)$  is the true Max-Cut. Conjecture: for every 3-regular graph  $G$  with  $n \geq 12$ ,  $\rho(G) \leq 1 - (1/10) \cdot ((\text{HCF}(G) - 1)/(n - 1))^{1/2}$ . A single 3-regular graph violating this strict inequality refutes the conjecture; in particular, any spectral-SOS 0.879-approximator would require  $\text{HCF}(G) < 1 + (10 \cdot 0.121)^2 \cdot (n - 1)$ .

## Rationale

The Laplacian spectrum is exactly the eigenvalue spectrum of the rank-1 degree-2 SOS moment-matrix surrogate  $B \propto v_1 v_1^T$  scaled by  $\lambda_1$ , and its 4-vs-2-norm ratio is the natural Boolean hypercontractive flatness invariant; a spectrum concentrated at the top ( $\text{HCF} \approx 1$ ) corresponds to a 'junta-like' moment matrix where the spectral relaxation is tight, while a spread spectrum ( $\text{HCF} \gg 1$ ) forces an integrality gap exactly because the dominant eigenvector loses bipartite alignment with  $G$ 's edges. Restricting to 3-regular graphs avoids the dense-graph ( $K_n$ ) and exact-cycle pathologies where the spectral bound is tight identically. The bridge between hypercontractive flatness of Laplacian spectra and Max-Cut integrality gaps appears to have <5 papers (Friedgut/Bourgain-style hypercontractivity has not been formally aimed at SOS Max-Cut moment matrices).

## Novelty

- Judge: NOVEL over 4 arXiv hits

Top hits consulted: - [1107.1630v3] On the Integrality Gap of the Subtour LP for the 1,2-TSP - [1207.5696v1] Beyond Max-Cut:  $\lambda$ -Extendible Properties Parameterized Above the Poljak-Turzík Bound - [2403.19550v2] Spectral gap for surfaces of infinite volume with negative curvature - [s2:1911.01411] Lifting sum-of-squares lower bounds: degree-2 to degree-4

## Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_67b46128.py", line 194, in <module>
    result = run_trial(seed)
              ^^^^^^^^^^^^^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_67b46128.py", line 148, in run_trial
    eigenvalues = sorted(matrix_eigenvalues(L))
                    ^^^^^^^^^^^^^^^^^^^^^^^^^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_67b46128.py", line 55, in matrix_eigenvalues
    Q[i][j] = Fraction(v[i], norm)
               ^^^^^^^^^^^^^^^^^
File "/usr/lib/python3.12/fractions.py", line 277, in __new__
    raise TypeError("both arguments should be "
TypeError: both arguments should be Rational instances
```

## Judge reasoning

The test crashed with a `TypeError` in `matrix_eigenvalues` before any of the 150 instances were evaluated, so no data exists to assess `support_fraction` or `min_margin`. Per rule 2, a crash yields **INCONCLUSIVE**. | next: Fix `matrix_eigenvalues` to use floating-point (`numpy.linalg.eigvalsh`) or coerce the norm to a Rational/sqrt-friendly type before re-running the 150-instance sweep.

---

## Free Cumulant Excess of Laplacian Spectrum Bounds Max-Cut SoS-2 Gap

- **Verdict:** INCONCLUSIVE
- **Bridge:** Free probability of Voiculescu-Speicher — non-crossing partition cumulants of the empirical spectral measure of a signed graph Laplacian, computed by Möbius inversion on the lattice  $\text{NC}(k)$ . This corner of operator-algebraic combinatorics has essentially no presence in CSP / SoS integrality-gap literature (an arXiv search for ‘free cumulant’ AND (‘Max-Cut’ OR ‘integrality gap’ OR ‘Sum-of-Squares’) returns no direct hits and <5 adjacent free-probability-meets-graph-spectra papers, all on Wigner-type ensembles). × Degree-2 Sum-of-Squares / Lasserre integrality gap for Max-Cut on random 3-regular graphs (the Goemans-Williamson / Delorme-Poljak regime, target of Schoenebeck-Khot-Vishnoi-Tulsiani SoS integrality-gap programs).
- **Recorded:** 2026-05-17 16:51 UTC
- **Entry ID:** c10c070e26bd

## Statement

For a 3-regular graph  $G$  on  $n \geq 8$  vertices, let  $\hat{L}_G = (D-A)/3$  be the normalized Laplacian,  $\mu_G$  its empirical spectral measure, and let  $\kappa_j(G)$  denote its  $j$ th free cumulant computed by Möbius inversion on  $\text{NC}(j)$  ( $\kappa_2 = m_2 - m_1^2$ ,  $\kappa_4 = m_4 - 4 m_3 m_1 - 2 m_2^2 + 10 m_2 m_1^2 - 5 m_1^4$ ). Define the free-Poisson excess  $\varepsilon(G) := \max(0, \kappa_4(G)/\kappa_2(G)^2 - 1)$  and the gap  $g(G) := \text{SDP}_2(G)/\text{MaxCut}(G)$ , where  $\text{SDP}_2(G) = (3n/8) \cdot (1 + \lambda_{\max}(\hat{L}_G - I))$  is the Delorme-Poljak spectral SoS-2 upper bound. CONJECTURE: for every 3-regular  $G$ ,  $(g(G) - 1) \cdot \log_2(n) \leq 4 \cdot (\varepsilon(G) + 1/n)$ . A single instance violating this bound, or any instance with  $g(G) > 1 + 4 \cdot (\varepsilon(G) + 1/n) / \log_2 n$ , falsifies the conjecture.

## Rationale

Free cumulants linearize free convolution and detect deviation from a Marchenko-Pastur / free-Poisson reference, so  $\varepsilon(G)$  measures exactly the bulk-spectrum ‘non-randomness’ that drives

Delorme-Poljak slack on sparse random Cayley-like graphs. Because SDP<sub>2</sub> is a strict function of the Laplacian spectrum while MaxCut depends on combinatorial cycle structure, any controlled bound on  $(\text{SDP}_2/\text{MaxCut} - 1)$  by a spectral free-probability quantity offers a CSP-instance witness rather than a truth-table predicate, avoiding the NATURAL\_PROOFS trap that killed the analogous Lasserre moment-rank attempt. The bridge is novel: free cumulants of graph Laplacians have been studied in random-matrix limits but never used as a finite- $n$  SoS-gap certifier.

## Novelty

- Judge: NOVEL over 0 arXiv hits

## Empirical Test

- exit code: 0, elapsed: 5.32s

```
conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'T(G)', 'metric_value': -2.125814583693911, 'instances_tested': 1, 'conjecture_holds': True}
TRIAL: {'metric_name': 'T(G)', 'metric_value': -2.125814583693911, 'instances_tested': 1, 'conjecture_holds': True}
TRIAL: {'metric_name': 'T(G)', 'metric_value': -2.25, 'instances_tested': 1, 'conjecture_holds': True}
TRIAL: {'metric_name': 'T(G)', 'metric_value': -2.060964047443681, 'instances_tested': 1, 'conjecture_holds': True}
TRIAL: {'metric_name': 'T(G)', 'metric_value': -2.060964047443681, 'instances_tested': 1, 'conjecture_holds': True}
TRIAL: {'metric_name': 'T(G)', 'metric_value': -2.0, 'instances_tested': 1, 'conjecture_holds': True}
TRIAL: {'metric_name': 'T(G)', 'metric_value': -2.125814583693911, 'instances_tested': 1, 'conjecture_holds': True}
TRIAL: {'metric_name': 'T(G)', 'metric_value': -2.125814583693911, 'instances_tested': 1, 'conjecture_holds': True}
TRIAL: {'metric_name': 'T(G)', 'metric_value': -2.0, 'instances_tested': 1, 'conjecture_holds': True}
TRIAL: {'metric_name': 'T(G)', 'metric_value': -2.25, 'instances_tested': 1, 'conjecture_holds': True}
TRIAL: {'metric_name': 'T(G)', 'metric_value': -2.060964047443681, 'instances_tested': 1, 'conjecture_holds': True}
TRIAL: {'metric_name': 'T(G)', 'metric_value': -2.25, 'instances_tested': 1, 'conjecture_holds': True}
TRIAL: {'metric_name': 'T(G)', 'metric_value': -2.060964047443681, 'instances_tested': 1, 'conjecture_holds': True}
TRIAL: {'metric_name': 'T(G)', 'metric_value': -2.060964047443681, 'instances_tested': 1, 'conjecture_holds': True}
RESULT: SUPPORTED mean=-2.1187349498381076 std=0.08924677387309159 support_fraction=1.0
```

## Judge reasoning

Although the test’s RESULT line reports SUPPORTED with  $T(G) \leq 0$  on all 150 instances, the critic’s challenge is well-founded: T values cluster near  $-2.1$ , indicating  $g(G) \approx 1$  on these small near-bipartite 3-regular graphs, so the bound is satisfied trivially rather than meaningfully tested. The pre-registered regime ( $n \leq 16$ ) does not probe the adversarial large- $n$  regime where the SDP-MaxCut gap actually opens. | next: Re-run on larger 3-regular graphs ( $n \in \{50, 100, 200, 500\}$ ) and include known integra

---

## Schatten-1/Schatten-4 Ratio Lower-Bounds AND-Function Communication

- **Verdict:** INCONCLUSIVE
- **Bridge:** Noncommutative  $L^p$  geometry — Schatten- $p$  norm ratios in the Pisier / Lust-Piquard framework, specifically the  $(S_1, S_4)$  stable-rank invariant  $\tau(M) = \log_2(|M| \{S_1\}^2 / |M| \{S_4\}^2)$ . This corner of operator-space theory is rarely applied to communication complexity beyond Forster’s operator-norm sign-rank bound; an arXiv search for ‘Schatten 4 norm’ AND ‘communication complexity’ returns <5 hits, all on unrelated invariants (e.g., spectral norm or  $\gamma_2$ ).  $\times$  Randomized two-party communication complexity of AND-functions  $M_g(x \wedge y)$  on  $\{0,1\}^n \times \{0,1\}^n$ , with the prototype  $g = \text{AND-of-NOTs}$  giving  $M = M_{\text{DISJ}_n}$  (Razborov 1992  $\Omega(n)$  regime). The conjecture instantiates the COMM\_DISJ tensor-invariant program and parallels the spectral-expansion mechanism underlying Tseitin-on-expander resolution width lower bounds (Ben-Sasson-Wigderson, Atserias-Bonet-Esteban).
- **Recorded:** 2026-05-17 17:24 UTC



- **Entry ID:** 87f5d4a9eef6

## Statement

For  $g: \{0,1\}^n \rightarrow \{0,1\}$ , let  $M_g \in \{0,1\}^{2^n \times 2^n}$  be the AND-function communication matrix  $M_g[x,y] = g(x \wedge y)$ . With  $\sigma_i =$  singular values of  $M_g$ , define  $\tau(M_g) := \log_2((\sum_i \sigma_i)^2 / (\sum_i \sigma_i^4)^{1/2}) = 2 \cdot \log_2(\|M_g\|_{S_1}) - \log_2(\|M_g\|_{S_4}^2)$ . Conjecture (A): for every AND-function,  $CC_R^{1/3}(M_g) \geq (1/8) \tau(M_g) - 2$ ; (B): for  $M_{DISJ_n}$  ( $g(z) = \prod_i (1 - z_i)$ ),  $\tau(M_{DISJ_n}) = n \cdot \log_2(5/\sqrt{7}) \approx 0.918n$ , exactly. A single AND-function  $M_g$  with  $\tau(M_g) > 8 \cdot D^{cc}(M_g) + 16$ , or any  $n$  with  $\tau(M_{DISJ_n}) < 0.7n$ , falsifies.

## Rationale

The DISJ matrix factors as a tensor power of the  $2 \times 2$  block  $(1,1;1,0)$  with golden-ratio spectrum  $\{\varphi, -1/\varphi\}$ , so its Schatten norms have closed forms  $(\|\cdot\|_{S_p})^p = (\varphi^p + \varphi^{(-p)})n$ , giving  $\tau = \Theta(n)$  by exact computation rather than spectral inequalities. The  $(S_1, S_4)$  ratio strengthens Forster's  $(S_2, S_\infty)$  ratio: it is more sensitive to mid-spectrum mass while remaining sub-multiplicative under tensor product, the same algebraic property that makes Tseitin-on-expanders amenable to width-via-expansion arguments. If proved, conjecture (A) yields a clean, calculable tensor invariant that matches Razborov's  $\Omega(n)$  bound on a wide AND-function class and would import noncommutative  $L^p$  techniques into proof-complexity-flavored CC lower bounds.

## Novelty

- Judge: NOVEL over 0 arXiv hits

## Empirical Test

- exit code: 1, elapsed: 0.20s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_624caab1.py", line 177, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_624caab1.py", line 145, in run_trial
    tau = compute_tau(matrix)
          ^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_624caab1.py", line 84, in compute_tau
    u, s, vt = svd(matrix)
               ^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_624caab1.py", line 55, in svd
    s[i] = matrix_norm([row[i] for row in b])
             ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_624caab1.py", line 33, in matrix_norm
    return math.sqrt(sum(sum(x**2 for x in row) for row in a))
                       ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_624caab1.py", line 33, in <genexpr>
    return math.sqrt(sum(sum(x**2 for x in row) for row in a))
                       ^^^^^^^^^^^^^^^^^^^^^
```

TypeError: 'float' object is not iterable

## Judge reasoning

The test crashed with a TypeError in the custom SVD/matrix\_norm implementation before any data was produced, so neither the support nor falsification criteria can be evaluated. | next: Fix the ma-

trix\_norm/SVD helper (use numpy.linalg.svd on the 2D matrix instead of the buggy nested-iteration norm) and rerun the n=4..8 sweep.

## Quiver Mutation Neighborhood Lower-Bounds ACC<sup>0</sup> by Sensitivity

- **Verdict:** INCONCLUSIVE
- **Bridge:** Cluster algebra theory of Fomin-Zelevinsky — finite-step quiver mutation neighborhoods (FZ 2002 ‘Cluster algebras I’; Keller’s mutation-class machinery). Mutations are non-ring combinatorial graph rewrites (arrow reversal at a vertex, path completion, 2-cycle cancellation) on directed multi-graphs; an arXiv search ‘quiver mutation’ AND (‘ACC<sup>0</sup>’ OR ‘circuit complexity’ OR ‘Boolean circuit’) returns no direct hits, with <5 adjacent papers using cluster algebras for combinatorial gadgets — never for circuit lower bounds. × ACC<sup>0</sup>[m] circuit size and depth for Boolean functions, in the Williams 2011 / Murray-Williams 2018 algorithmic-method regime targeting NEXP  $\not\subseteq$  ACC<sup>0</sup> and MOD<sub>q</sub> lower bounds for q coprime to m. The invariant is a STRUCTURAL property of the circuit DAG (not of its truth table), aligning with Williams-style structural exploitations of small-circuit structure.
- **Recorded:** 2026-05-17 17:51 UTC
- **Entry ID:** c7f898dc68a4

### Statement

For an ACC<sup>0</sup>[m] circuit C with N gates and depth d computing  $f: \{0,1\}^n \rightarrow \{0,1\}$  with average sensitivity  $I(f) = E_x[\#\{i: f(x \oplus e_i) \neq f(x)\}]$ , let Q(C) be the multi-quiver whose vertices are circuit nodes (inputs and gates) and whose arc multiplicity  $i \rightarrow g$  equals the number of times node i feeds gate g, with 2-cycles cancelled. Define  $NbhdSize(C) := \#\{\text{isomorphism classes of } \mu_v(Q(C)) : v \in V(Q(C))\}$ , where  $\mu_v$  is the Fomin-Zelevinsky mutation at v (reverse all arrows at v; for each path  $i \rightarrow v \rightarrow j$  add an arrow  $i \rightarrow j$  with multiplicity  $\text{mult}(i \rightarrow v) \cdot \text{mult}(v \rightarrow j)$ ; cancel resulting 2-cycles), and isomorphism is detected by Weisfeiler-Lehman color refinement on the labeled multi-digraph. Conjecture:  $NbhdSize(C) \geq \lceil I(f)/(4(d+1)) \rceil$  for every ACC<sup>0</sup>[m] circuit; a single C with  $NbhdSize(C) < \lceil I(f)/(4(d+1)) \rceil$  refutes the conjecture.

### Rationale

Quiver mutation is a non-commutative, non-ring graph rewrite that cancels 2-cycles — an operation that does not lift coherently to polynomial extensions of the Boolean ring, placing the invariant outside the Aaronson-Wigderson algebrization frame that has tripped past GCT-flavoured ACC<sup>0</sup> attempts. NbhdSize is a structural circuit invariant rather than a truth-table predicate, so the Razborov-Rudich largeness/usefulness clauses do not apply and the natural-proofs barrier is sidestepped (the same f admits circuits with very different NbhdSize). Heuristically, high-sensitivity targets force the gate-multi-quiver to be mutation-rigid (many non-isomorphic 1-step neighbours), because low-rigidity quivers correspond to symmetric, low-sensitivity computation — giving a Williams-style structural-versus-functional tension.

### Novelty

- Judge: NOVEL over 0 arXiv hits

### Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6aabdc6d.py", line 160, in <module>
    result = run_trial(seed)
```

```

^^^^^^^^^^^^^^^^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6aabdc6d.py", line 136, in run_trial
    mutated_quiver = mutate_quiver(quiver, vertex)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6aabdc6d.py", line 57, in mutate_quiver
    for u in quiver:
RuntimeError: dictionary changed size during iteration

```

## Judge reasoning

The test crashed with a `RuntimeError` (dictionary changed size during iteration) in `mutate_quiver` before any trials produced data, so neither the support nor falsification condition can be evaluated.  
| next: Fix `mutate_quiver` to iterate over a snapshot (e.g., `list(quiver.items())`) or build the mutated quiver in a fresh dict, then rerun the 90-trial pre-registered protocol.

---

## Bureau Trace Defect of Clause Braid Lower-Bounds Tree-Resolution Steps

- **Verdict:** INCONCLUSIVE
- **Bridge:** Braid-group representation theory — the unreduced Bureau representation  $\rho_t: B_n \rightarrow GL_n(\mathbb{Z}[t, t^{-1}])$  specialised at  $t = -1$ , where Bureau factors through the integer symplectic group  $Sp_{2g}(\mathbb{Z})$  (Bureau 1936; Squier 1984; Birman 1974), and the trace defect  $\omega(\beta) := |n - \text{tr}(\rho_{-1}(\beta))|$ . This corner of knot/braid theory has essentially no presence in proof complexity (arXiv search ‘Bureau representation’ AND (‘Frege’ OR ‘resolution refutation’ OR ‘proof complexity’) returns no direct hits; <5 adjacent papers). × Tree-like Resolution / DPLL refutation step count for unsatisfiable 3-CNFs, viewed as a Frege-depth proxy in the Cook-Reckhow-Krajíček bounded-arithmetic framework where  $\neg F$  is  $\Sigma^b_0$  and  $\log_2$  of the canonical tree size controls  $V^0$ -witnessing complexity (Beame-Pitassi 2001; Ben-Sasson-Wigderson 2001).
- **Recorded:** 2026-05-17 18:27 UTC
- **Entry ID:** b455093f5476

## Statement

For an unsat 3-CNF  $F$  on  $n$  variables with  $m$  clauses, construct a braid word  $\beta(F) \in B_n$  by reading clauses  $C \in F$  in lex order: for each  $C$  with literals  $l_1, l_2, l_3$  sorted by  $|var|$ , append generators  $\sigma_{\{|l_1|\}^{\text{sgn}(l_1) \cdot \text{sgn}(l_2)}} \cdot \sigma_{\{|l_2|\}^{\text{sgn}(l_2) \cdot \text{sgn}(l_3)}}$  ( $\sigma_k$  for exponent  $+1$ ,  $\sigma_k^{-1}$  for exponent  $-1$ , with  $\sigma_k$  acting between strands  $k$  and  $k+1$ ,  $k \leq n-1$ ; pairs that straddle strand  $n-1$  are skipped). Let  $M(F) = \rho_{-1}(\beta(F))$  be the unreduced Bureau matrix product specialised at  $t = -1$  (so each generator is an integer  $n \times n$  matrix with entries in  $\{-1, 0, 1, 2\}$ ), and  $\omega(F) := |n - \text{tr}(M(F))|$ . Conjecture: there is an absolute constant  $C \geq 1/4$  such that the canonical lex-DPLL refutation backtrack count  $t(F)$  satisfies  $4 \cdot (t(F)^2 + 1) \geq \omega(F)$  for every unsat 3-CNF  $F$ . A single unsat  $F$  with  $4 \cdot (t(F)^2 + 1) < \omega(F)$  falsifies the conjecture.

## Rationale

At  $t = -1$  the Bureau representation lands inside  $Sp_{2g}(\mathbb{Z})$ , so the matrix product realises clause-pair conflicts as integer symplectic transvections and the trace defect quantifies how entangled the resulting symplectic flow becomes — a relational, non-commutative invariant of the SYNTACTIC clause ordering that is invisible to width or expansion alone. Any tree-Resolution refutation must split on every variable participating in an irreducible conflict, so a highly entangled clause braid should force backtracking. The invariant is structural (CNF syntax, not truth-table), sidestepping NATURAL\_PROOFS; braids are non-abelian and do not lift to polynomial extensions of the Boolean ring, sidestepping ALGEBRIZATION; the construction uses no diagonalisation, sidestepping RELATIVIZATION.

## Novelty

- Judge: NOVEL over 2 arXiv hits

Top hits consulted: - [s2:68cab25640a4debd147cd412a08597d352b33] Refutation of a Proof Complexity Approach to Maxsat Resolution-Based Systems - [3506702] On Proof Complexity of Resolution over Polynomial Calculus

## Empirical Test

- exit code: 0, elapsed: 0.04s

```
alue": 0, "instances_tested": 0, "conjecture_holds": false, "counterexample": "No valid instances gen
TRIAL: {"seed": 463, "metric_name": "omega", "metric_value": 0, "instances_tested": 0, "conjecture_ho
TRIAL: {"seed": 503, "metric_name": "omega", "metric_value": 0, "instances_tested": 0, "conjecture_ho
TRIAL: {"seed": 547, "metric_name": "omega", "metric_value": 0, "instances_tested": 0, "conjecture_ho
TRIAL: {"seed": 593, "metric_name": "omega", "metric_value": 0, "instances_tested": 0, "conjecture_ho
TRIAL: {"seed": 631, "metric_name": "omega", "metric_value": 0, "instances_tested": 0, "conjecture_ho
TRIAL: {"seed": 677, "metric_name": "omega", "metric_value": 0, "instances_tested": 0, "conjecture_ho
TRIAL: {"seed": 727, "metric_name": "omega", "metric_value": 0, "instances_tested": 0, "conjecture_ho
TRIAL: {"seed": 773, "metric_name": "omega", "metric_value": 0, "instances_tested": 0, "conjecture_ho
TRIAL: {"seed": 821, "metric_name": "omega", "metric_value": 0, "instances_tested": 0, "conjecture_ho
TRIAL: {"seed": 877, "metric_name": "omega", "metric_value": 0, "instances_tested": 0, "conjecture_ho
TRIAL: {"seed": 929, "metric_name": "omega", "metric_value": 0, "instances_tested": 0, "conjecture_ho
RESULT: FALSIFIED counterexample="No valid instances generated" first_failing_seed=11
```

## Judge reasoning

All 30 seeds produced zero accepted instances ('No valid instances generated'), so the conjecture was never actually evaluated; the RESULT line's 'FALSIFIED' label reflects a generator failure rather than a genuine counterexample. Pre-registered acceptance threshold ( $n_{\text{accepted}} \geq 400$ ) was not remotely met. | next: Fix the instance generator (e.g., loosen acceptance, verify DPLL unsat check, ensure clauses produce non-skipped Bureau generators at  $\alpha=6$ ,  $n \in \{8..16\}$ ) and rerun to obtain a non-empty sam

---

## Euler Characteristic of Gate-Conflict Independence Complex Bounds $\text{ACC}^0[m]$ Size for $\text{MOD}_q$

- **Verdict:** INCONCLUSIVE
- **Bridge:** Topological combinatorics — reduced Euler characteristic  $\tilde{\chi}(\text{Ind}(G))$  of the independence complex of a finite graph  $G$ , computed as the independence-polynomial value  $I(G;-1)$  (Kozlov 2008 'Combinatorial Algebraic Topology'; Engström 2009 'Complexes of directed trees and independence complexes'; Stanley-Reisner / Alexander duality). This corner of combinatorial algebraic topology has essentially no presence in circuit complexity (arXiv 'independence complex' AND (' $\text{ACC}^0$ ' OR 'circuit complexity' OR 'Williams algorithmic method') returns no direct hits; <5 adjacent monotone-circuit-matroid papers).  $\times \text{ACC}^0[m]$  circuit size for  $\text{MOD}_q$  on  $n$  inputs with  $q$  coprime to  $m$  (focus  $q=3$ ,  $m=2$ ), in the Williams 2011 / Murray-Williams 2018 algorithmic-method regime targeting  $\text{NEXP} \not\subseteq \text{ACC}^0$ . The invariant is a STRUCTURAL property of the circuit DAG (gate-conflict graph), not of the truth table, so it is shielded from the Razborov-Rudich natural-proofs barrier.
- **Recorded:** 2026-05-17 19:02 UTC
- **Entry ID:** cdef279c90bb

## Statement

For an  $\text{ACC}^0[m]$  circuit  $C$  of depth  $d$  and size  $s$  on  $n$  inputs, let  $G_C$  be the gate-conflict graph: vertices = non-input gates;  $(g,h)$  is an edge iff the input-variable supports  $\text{supp}(g)$ ,  $\text{supp}(h) \subseteq [n]$  satisfy  $\text{supp}(g) \cap \text{supp}(h) \neq \emptyset$ . Let  $\tau(C) := |I(G_C; -1)|$  where  $I(G; x) = \sum_{\{S \text{ independent in } G\}} x^{|S|}$ . Conjecture: every depth- $d$   $\text{ACC}^0[m]$  circuit  $C$  that  $\varepsilon$ -approximates  $\text{MOD}_q$  on  $n$  inputs ( $q$  coprime to  $m$ ,  $\varepsilon \leq 1/4$ ,  $q \geq 3$ ) satisfies  $\tau(C) \geq 2^{\{n/(10 \cdot d^2)\}} - 1$ . A single  $(C, q, \varepsilon)$  instance with  $\tau(C) < 2^{\{n/(10 \cdot d^2)\}} - 1$  yet  $\text{agreement}(C, \text{MOD}_q) \geq (1-\varepsilon) \cdot 2^n$  refutes it.

## Rationale

Razborov-Smolensky shows that approximating  $\text{MOD}_q$  in  $\text{AC}^0[p]$  forces large size via polynomial-method incompatibility; the independence complex of the gate-conflict graph categorifies the variable-overlap pattern that polynomial-degree bounds exploit, with  $\tilde{\chi}$  measuring ‘how rigidly’ supports must intersect to express many distinct minterms. Because  $\tau(C)$  depends only on the wiring DAG (not the truth table), the invariant cannot be ‘large + useful’ in the Razborov-Rudich sense, dodging natural proofs; because it is built from set intersections and Möbius sign sums (not ring operations on tt-extensions), it does not algebrize in the Aaronson-Wigderson sense. The bound  $2^{\{n/(10 \cdot d^2)\}}$  aligns with Smolensky’s  $\sqrt{n}$  degree barrier propagated through  $d$  layers.

## Novelty

- Judge: NOVEL over 9 arXiv hits

Top hits consulted: - [1904.05483v2] Parallels Between Phase Transitions and Circuit Complexity? - [1606.05050v1] Proof Complexity Lower Bounds from Algebraic Circuit Complexity - [9906008v2] A Lower Bound on the Average-Case Complexity of Shellsort - [2304.04810v2] Chain algebras of finite distributive lattices - [1110.6876v4] Lower bounds for topological complexity

## Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_cbad24de.py", line 135, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_cbad24de.py", line 103, in run_trial
    truth_table = compute_truth_table(circuit, n)
                  ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_cbad24de.py", line 45, in compute_truth
    output = input_values[gate_inputs[0]] ^ input_values[gate_inputs[1]]
             ~~~~~^
```

IndexError: list index out of range

## Judge reasoning

The test crashed with an `IndexError` before producing any data, so the pre-registered support condition could not be evaluated and no qualifying instances were assessed. | next: Fix the circuit-evaluation bug (guard `gate_inputs` indexing against gates with fewer than 2 inputs, or generate circuits whose gate fan-in matches the evaluator) and rerun the trial sweep.

---

## Tamari Rotation Rank of Canonical DT Lower-Bounds XOR-Lifted F2-Rank

- Verdict: INCONCLUSIVE

- **Bridge:** Tamari lattice combinatorics — the order-theoretic Pallo rank function  $\text{rrd}(T) = \sum$  over internal nodes  $v$  of  $|L(v)|$  (size of  $v$ 's left-subtree leaf set), in the Sleator-Tarjan-Thurston / Pallo-Bonnin / Pournin / Chapoton tradition on Catalan-counted binary trees. An arXiv search for 'Tamari lattice' AND ('communication complexity' OR 'lifting theorem' OR 'decision tree complexity') returns no direct hits and <5 adjacent papers — this corner of Catalan/order combinatorics has essentially no presence in CC / lifting literature.  $\times$  F2-rank of the XOR-lifted communication matrix  $M_{\{f \text{ xor}\}}$  for Boolean  $f: \{0,1\}^n \rightarrow \{0,1\}$ , defined by  $M[a,b] = f(a \text{ XOR } b)$ , in the Raz-McKenzie / Goos-Pitassi-Watson lifting framework.  $\log_2 \text{rank}_{\{F2\}}(M)$  is the canonical algebraic lower bound for deterministic two-party  $\text{CC}^D(M_{\{f \text{ xor}\}})$ , the XOR-function CC studied since Lee-Shraibman, Hatami-Hosseini-Lovett, and the Log-Rank Conjecture.
- **Recorded:** 2026-05-17 19:17 UTC
- **Entry ID:** b045e08ff6c7

## Statement

For a Boolean function  $f: \{0,1\}^n \rightarrow \{0,1\}$ , let  $T_{f^*}$  be the canonical minimum-leaf decision tree obtained by a deterministic DP on partial assignments: at each restriction state, branch on the smallest-index variable whose two children have strictly smaller leaf totals, with lex tie-breaking on serialized (left-shape, right-shape) tuples. Let  $\text{rrd}(T_f) = \sum$  over internal nodes  $v$  of  $|L(v)|$ , the Pallo right-rotation rank in the Tamari lattice. Define  $M = M_{\{f \text{ xor}\}}$ , the  $2^n$  by  $2^n$  matrix  $M[a,b] = f(a \text{ XOR } b)$ . Conjecture: there is a universal constant  $c \geq 1/4$  such that for every  $f$  with  $\text{rrd}(T_f) \geq 4$ ,  $\log_2(\text{rank}_{\{F2\}}(M)) \geq c * \log_2(1 + \text{rrd}(T_f))$ . A single  $f$  with  $\text{rrd}(T_f) \geq 4$  but  $\log_2(\text{rank}_{\{F2\}}(M)) < 0.25 * \log_2(1 + \text{rrd}(T_{f^*}))$  refutes the conjecture.

## Rationale

Raz-McKenzie / GPW lifting passes the STRUCTURE of a decision tree  $T_f$  into the rectangle/block structure of  $M_{\{f \text{ o } g\}}$ ; the Pallo rank  $\text{rrd}(T_f)$  is the most natural Catalan-order measure of how 'right-comb-asymmetric' that block structure is, in a regime never previously connected to CC. The invariant is intrinsic to the chosen minimum tree  $T_f$  (a non-polynomial selection over restrictions), so it is not a polynomial in truth-table entries and the construction does not extend to polynomial extensions of the Boolean ring — sidestepping algebrization. The bridge respects LIFTING's safe directions (no relativization, no Karp-Lipton) and targets a CC-rank object, not a P/poly truth-table predicate, so natural-proofs does not bite.

## Novelty

- Judge: NOVEL over 2 arXiv hits

Top hits consulted: - [s00037-025-00276-5] Lifting Dichotomies - [s2:7250a91643b33e65c107da27b0c02900b7] Lifting Theorems for Equality

## Empirical Test

- exit code: 0, elapsed: 0.11s

```
ces_tested": 1, "conjecture_holds": true, "counterexample": ""}
TRIAL: {"seed": 503, "metric_name": "log2(rank_F2(M)) / log2(1 + rrd(T_f*))", "metric_value": 0.79295}
TRIAL: {"seed": 547, "metric_name": "log2(rank_F2(M)) / log2(1 + rrd(T_f*))", "metric_value": 0.73395}
TRIAL: {"seed": 593, "metric_name": "log2(rank_F2(M)) / log2(1 + rrd(T_f*))", "metric_value": 0.99628}
TRIAL: {"seed": 631, "metric_name": "log2(rank_F2(M)) / log2(1 + rrd(T_f*))", "metric_value": 0.83023}
TRIAL: {"seed": 677, "metric_name": "log2(rank_F2(M)) / log2(1 + rrd(T_f*))", "metric_value": 0.73395}
TRIAL: {"seed": 727, "metric_name": "log2(rank_F2(M)) / log2(1 + rrd(T_f*))", "metric_value": 0.99119}
TRIAL: {"seed": 773, "metric_name": "log2(rank_F2(M)) / log2(1 + rrd(T_f*))", "metric_value": 0.99119}
TRIAL: {"seed": 821, "metric_name": "log2(rank_F2(M)) / log2(1 + rrd(T_f*))", "metric_value": 0.85577}
TRIAL: {"seed": 877, "metric_name": "log2(rank_F2(M)) / log2(1 + rrd(T_f*))", "metric_value": 0.79295}
TRIAL: {"seed": 929, "metric_name": "log2(rank_F2(M)) / log2(1 + rrd(T_f*))", "metric_value": 0.79295}
```

RESULT: SUPPORTED mean=0.8822973003045711 std=0.09969545211010329 support\_fraction=1.0

## Judge reasoning

Random-seed sampling reports mean ratio 0.88 with min 0.73, but the critic identifies a structured adversarial case (parity function) where  $\text{rrd} \sim n \cdot 2^n$  while F2-rank of the XOR-lift is 2, yielding ratio  $\approx 0.166 < 0.25$  for  $n=4$  — not probed by the random sample. Per the strict rule, an unrefuted CHALLENGE downgrades the verdict. | next: Explicitly evaluate the conjecture on the parity function  $f(x)=x_1 \oplus \dots \oplus x_n$  for  $n$  in  $\{4,5,6,7\}$  and other low-F2-rank structured functions (e.g., affine, inner product)

---

## Immanant Log-Variance Separates Random Matrices from Det-Padded Blocks

- **Verdict:** INCONCLUSIVE
- **Bridge:** Symmetric-group representation theory: the Specht-character immanant vector  $\text{imm\_lambda}(M)=\sum_{\sigma} \chi^{\text{lambda}}(\sigma) \prod_i M[i,\sigma(i)]$  as a partition-indexed structural decomposition of a matrix function (Stembridge 1991; Stanley EC2 Ch.7; Greene’s immanant-positivity program). The invariant is RELATIONAL (vector-valued across  $p(n)$  partitions, not a single-number truth-table functional) and distinct from the blacklisted positivity-width attempt: the present invariant is the LOG-MAGNITUDE VARIANCE across the full Specht spectrum on PADDING-vs-RANDOM ensembles, an  $\exp(n)$ -time meta-complexity-style discriminator, not a natural-proof-style poly-time invariant.  $\times$  Meta-complexity in the GCT\_DET\_PERM regime: distinguishing  $K^t$ -random Boolean matrices from low- $K^t$  / orbit-closure-of- $\det_m^{\{\text{padded}\}}$  matrices via a super-polynomial structural test, in the spirit of Hirahara’s GapMCSP and the Mulmuley-Sohoni orbit-closure separation of  $\text{perm}_n$  from  $\det_m^{\{\text{poly}\}}$ . The invariant  $V(M)$  is a candidate meta-complexity measure that lower-bounds compressibility under linear/block padding without being itself in P.
- **Recorded:** 2026-05-17 19:47 UTC
- **Entry ID:** 88defc39f508

## Statement

For  $M$  in  $\{0,1\}^{\{n \times n\}}$  define  $I_{\text{lambda}}(M) := \sum_{\sigma \in S_n} \chi^{\text{lambda}}(\sigma) \prod_i M[i,\sigma(i)]$  using the Murnaghan-Nakayama character table, and set  $V(M) := \text{Var}_{\{\text{lambda} \in P(n)\}} (\log_2(1 + |I_{\text{lambda}}(M)|))$ . Conjecture: for each  $n$  in  $\{5,6,7\}$  there exist absolute constants  $c1 \geq 4c2 > 0$  such that (i) over uniform  $M$  in  $\{0,1\}^{\{n \times n\}}$ ,  $V(M) \geq c1(\log_2 n)^2$  on  $\geq 27$  of 30 seeds, while (ii) over the det-padded ensemble  $M_{\text{pad}} = \text{block-diag}(M', J_{\{n-m,n-m\}})$  with  $m = \text{ceil}(n/2)$ ,  $M'$  uniform in  $\{0,1\}^{\{m \times m\}}$  and  $J$  all-ones,  $V(M_{\text{pad}}) \leq c2(\log_2 n)^2$  on  $\geq 27$  of 30 seeds. A single  $(n, \text{seed})$  pair where  $V(\text{uniform}) < 4V(\text{padded})$  on independent draws falsifies the conjecture.

## Rationale

Padding a smaller block by an all-ones submatrix forces the immanant sum to factor through the permanent of the J-block, collapsing the Specht spectrum onto partitions adjacent to  $(n)$  and  $(1^n)$  and zeroing most middle-lambda components by row/column character cancellation. Uniform random Boolean matrices instead distribute i.i.d. signed contributions across all  $p(n)$  Specht coordinates, giving  $\Theta(\sqrt{n!})$  magnitudes per lambda and high log-variance. The bridge from partition-indexed Specht variance to the Mulmuley-Sohoni padding question parallels how stabilizer-codimension invariants stratify det vs perm: padding by structureless blocks erases the representation-theoretic spread that random matrices exhibit.

## Novelty

- Judge: NOVEL over 6 arXiv hits

Top hits consulted: - [1005.0979v1] Supersymmetry in Random Matrix Theory - [0911.5645v1] Non-Hermitian Random Matrix Ensembles - [0904.1432v2] Applications of random matrix theory to condensed matter and optical physics - [1611.00827v2] Geometric complexity theory and matrix powering - [1911.03990v1] Implementing geometric complexity theory: On the separation of orbit closures via symmetries

## Empirical Test

- exit code: 0, elapsed: 6.81s

```
_name': 'log_variance_ratio', 'metric_value': 0.0, 'instances_tested': 3, 'conjecture_holds': True,
TRIAL: {'metric_name': 'log_variance_ratio', 'metric_value': 0.0, 'instances_tested': 3, 'conjecture
TRIAL: {'metric_name': 'log_variance_ratio', 'metric_value': 0.0, 'instances_tested': 3, 'conjecture
TRIAL: {'metric_name': 'log_variance_ratio', 'metric_value': 0.0, 'instances_tested': 3, 'conjecture
TRIAL: {'metric_name': 'log_variance_ratio', 'metric_value': 0.0, 'instances_tested': 3, 'conjecture
TRIAL: {'metric_name': 'log_variance_ratio', 'metric_value': 0.0, 'instances_tested': 3, 'conjecture
TRIAL: {'metric_name': 'log_variance_ratio', 'metric_value': 0.0, 'instances_tested': 3, 'conjecture
TRIAL: {'metric_name': 'log_variance_ratio', 'metric_value': 0.0, 'instances_tested': 3, 'conjecture
TRIAL: {'metric_name': 'log_variance_ratio', 'metric_value': 0.0, 'instances_tested': 3, 'conjecture
TRIAL: {'metric_name': 'log_variance_ratio', 'metric_value': 0.0, 'instances_tested': 3, 'conjecture
TRIAL: {'metric_name': 'log_variance_ratio', 'metric_value': 0.0, 'instances_tested': 3, 'conjecture
TRIAL: {'metric_name': 'log_variance_ratio', 'metric_value': 0.0, 'instances_tested': 3, 'conjecture
TRIAL: {'metric_name': 'log_variance_ratio', 'metric_value': 0.0, 'instances_tested': 3, 'conjecture
TRIAL: {'metric_name': 'log_variance_ratio', 'metric_value': 0.0, 'instances_tested': 3, 'conjecture
TRIAL: {'metric_name': 'log_variance_ratio', 'metric_value': 0.0, 'instances_tested': 3, 'conjecture
RESULT: SUPPORTED mean=0.0000 std=0.0000 support_fraction=1.0000
```

## Judge reasoning

Every trial reports `metric_value=0.0` with `std=0.0` and only 3 instances per seed rather than the pre-registered 30, so the two-sided inequality ( $V_{\text{uniform}} \geq c_1(\log n)^2$  and  $V_{\text{padded}} \leq c_2(\log n)^2$ ) is never actually evaluated. The degenerate identically-zero 'log\_variance\_ratio' indicates the test does not compute the conjectured quantity, so SUPPORTED cannot be trusted. | next: Reimplement the test to explicitly compute  $V(\text{uniform})$  and  $V(\text{padded})$  per seed via the Murnaghan-Nakayama character ta

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## Hypercontractive Term-Support Stable Rank Bounds Monotone Clique DNF

- **Verdict:** INCONCLUSIVE
- **Bridge:** Fourier analysis / hypercontractivity — Bonami-Beckner noise-operator Gram kernels on DNF-term SUPPORT sets, evaluated through the stable rank  $\sigma(K) = (\sum \lambda_i)^2 / \sum \lambda_i^2$  of the resulting term-correlation matrix, in the Kahn-Kalai-Linial / Friedgut / Mossel tradition but used as a SYNTACTIC DNF invariant (a property of the formula, not its truth table) — an angle with essentially no presence in monotone-circuit lower-bound literature outside Razborov-approximator analogies × Monotone DNF / formula complexity for the k-CLIQUE indicator on v-vertex graphs (Razborov 1985 regime, target measure  $\mu$  on monotone DNFs satisfying (i) sub-modular under conjunction, (ii)  $O(\log n)$  for poly-size DNFs, (iii)  $\Omega(n)/\Omega(v)$  for the k-CLIQUE indicator), accessed through the canonical minterm DNF where each term is an edge-set of a k-clique
- **Recorded:** 2026-05-17 20:14 UTC
- **Entry ID:** fad53ef695ab



## Statement

For a monotone DNF  $F = \bigvee_{i=1}^s T_i$  on  $N$  variables with each term  $T_i = \bigwedge_{k \in S_i} x_k$  indexed by its support set  $S_i \subseteq [N]$ , define the Bonami-Beckner term-correlation Gram matrix  $K(F) \in \mathbb{R}^{s \times s}$  at noise parameter  $\rho=1/2$  by  $K(F)[i,j] := 2^{-|S_i|-|S_j|} \cdot (3/2)^{|\{S_i \cap S_j\}|}$ , and set  $\mu(F) := \log_2((\sum_i \lambda_i(K))^2 / \sum_i \lambda_i(K)^2)$ , the log of the stable rank of  $K(F)$ . Then (i) for every monotone DNF  $F$  on  $N$  variables with  $s$  terms,  $\mu(F) \leq 4 \cdot \log_2(2N+s)$ ; (ii) for the  $k$ -CLIQUE $_v$  indicator on  $N=C(v,2)$  edge variables with  $k=\lceil \log_2 v \rceil$ , in its canonical minterm DNF (one term per  $k$ -clique of  $K_v$ , support = edge set of the clique),  $\mu(F_{\{k\text{-CLIQUE}_v\}}) \geq v/4$ ; (iii)  $\mu$  is submodular under conjunction:  $\mu(F \wedge G) \leq \mu(F) + \mu(G)$  for any monotone DNFs  $F, G$  (writing  $F \wedge G$  in its distributed-out form  $\bigvee_{i,j} T_i \wedge T'_j$ ). A single instance violating (i), (ii), or (iii) refutes the conjecture.

## Rationale

The Bonami-Beckner kernel at  $\rho=1/2$  admits a per-coordinate rank-2 factorization ( $f(a,b)=2^{\{-a-b\}(3/2)}\{ab\}$  with  $\det>0$ ), so  $K(F)=\bigotimes_k K_k$  restricted to the chosen supports — a hypercontractive Gram structure that controls effective dimensionality of the bottom monomial layer; poly-size DNFs cannot escape the trivial stable-rank  $\leq s \leq \text{poly}(N)$  bound, giving (i). The rigid intersection lattice of  $k$ -clique edge sets (two  $k$ -cliques sharing  $j$  vertices share  $C(j,2)$  edges, partitioning the  $s^2=C(v,k)^2$  pairs into  $k+1$  strongly-regular orbits) injects  $v$  many independent spectral components into  $K$ , driving stable rank to  $\geq 2^{\{v/4\}}$ . Submodularity under conjunction follows from tensor structure on disjoint variables (equality) and Cauchy-Schwarz contraction of merged Gram blocks on shared variables; crucially  $\mu$  is a syntactic invariant of the DNF, dodging NATURAL\_PROOFS since computing it requires the formula, not the truth table.

## Novelty

- Judge: NOVEL over 9 arXiv hits

Top hits consulted: - [1806.02734v3] Spectral lower bounds for the orthogonal and projective ranks of a graph - [1801.08709v2] Adaptive Lower Bound for Testing Monotonicity on the Line - [1110.6594v1] Around Operator Monotone Functions - [2501.12063v2] Gram-like matrix preserving extensions and completions of noncommutative polynomials - [2510.18132v1] Beurling Nyman Geometry and Gram Matrix Structure, Ladder Density and Polynomial Decay via Mellin Smoothing

## Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d4d4e1b2.py", line 166, in <module>
 result = run_trial(seed)
 ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d4d4e1b2.py", line 107, in run_trial
 mu = compute_mu(K)
 ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d4d4e1b2.py", line 56, in compute_mu
 eigenvalues = matrix_eigenvalues(K)
 ^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d4d4e1b2.py", line 33, in matrix_eigenvalues
 raise ValueError("Matrix too large for this implementation")
```

ValueError: Matrix too large for this implementation

## Judge reasoning

The test crashed with a ValueError ('Matrix too large for this implementation') before any instance could be evaluated, so no support\_fraction was produced and the pre-registered crite-

rion is neither met nor falsified. | next: Replace the bespoke `matrix_eigenvalues` routine with `numpy.linalg.eigvalsh` (or `scipy`) so the Gram matrices for Families A/B/C can be diagonalized at the required sizes, then rerun the 270+ instance battery.

## S-Transform Free-Mult Defect Lower-Bounds DISJ Communication

- **Verdict:** INCONCLUSIVE
- **Bridge:** Free probability — Voiculescu’s S-transform and free multiplicative convolution  $\boxtimes$  of compactly supported measures on  $\mathbb{R}_+$  (Voiculescu 1987; Bercovici-Voiculescu 1992; Haagerup-Schultz 2007). Multiplicative free convolution is distinct from additive (R-transform) free convolution previously attempted on graph Laplacians; an arXiv search for ‘S-transform’ AND (‘communication complexity’ OR ‘sign rank’ OR ‘discrepancy method’) returns 0 direct hits with <5 adjacent papers, all on Wigner-type ensembles in QIT, never on Boolean communication matrices.  $\times$  Randomized two-party communication complexity  $CC_R(M)$  of DISJOINTNESS and other Boolean matrices  $M \in \{0,1\}^{\{N \times N\}}$  (Razborov 1992  $\Omega(n)$  regime), with  $N = 2^n$ ; the  $\tau(M)$  tensor-invariant program for COMM\_DISJ.
- **Recorded:** 2026-05-17 20:26 UTC
- **Entry ID:** 6c8daf2ab19d

### Statement

For Boolean  $M \in \{0,1\}^{\{N \times N\}}$  let  $Q_M := (1/N) \cdot M^T M$  and let  $\sigma_M$  be its empirical spectral measure, normalized to unit total mass and unit first moment ( $m_1=1$ ) by rescaling. Compute moments  $m_k = (1/N) \text{tr}((Q_M/m_1)^k)$  for  $k=1..K$  with  $K=\lceil \log_2 N \rceil + 2$ ; form the truncated  $\psi$ -series  $\psi(z) = \sum m_k z^k$ , invert it to  $\chi(z) = \psi^{-1}(z)$  by Lagrange inversion, set  $S_M(z) = (1+z) \cdot \chi(z)/z$ , and obtain the moments  $\{\mu_k\}$  of the free multiplicative self-convolution  $\sigma_M \boxtimes \sigma_M$  from  $S_{\{\boxtimes\}}(z) = S_M(z)^2$  by re-inverting. Define the free-mult defect  $\delta(M) := |\mu_2 - 2 \cdot m_2 - 1| / (1 + m_2)$ . Conjecture: there is an absolute constant  $c \in [1/64, 1/4]$  such that (i) for every Boolean  $M$ ,  $CC_R(M) \geq c \cdot \log_2(N) \cdot \delta(M)$ ; (ii)  $\delta(M_{\text{DISJ}_n}) \geq 1/8$  for all  $n \geq 3$ . A single Boolean  $M$  with  $U(M) < c \cdot \log_2(N) \cdot \delta(M)$  (where  $U$  is any certified CC upper bound), or a single  $n \geq 3$  with  $\delta(M_{\text{DISJ}_n}) < 1/8$ , refutes the conjecture.

### Rationale

Multiplicative free convolution  $\sigma \boxtimes \sigma$  encodes asymptotic FREENESS between the row-space and column-space spectral projections of  $M$ ; deviation from the freeness identity  $\mu_2 = 2m_2 + 1$  detects multiplicative coupling between input partitions, which is exactly the structural property Razborov exploits to force long bipartite rectangles in DISJ. Unlike additive R-transform invariants (previously tried on graph Laplacians), the S-transform is intrinsically multiplicative and respects the row  $\times$  column tensor structure of CC matrices. The bridge is barrier-shielded: Lagrange inversion is non-polynomial in the entries, the invariant is RELATIONAL (a comparison between two measures, not a single truth-table functional), CC lower bounds are not subject to the natural-proofs barrier, and the construction uses no ring-only operations on a Boolean ring extension, so it does not algebrize.

### Novelty

- Judge: NOVEL over 13 arXiv hits

Top hits consulted: - [2004.07191v1] Cauchy-Stieltjes kernels families and free multiplicative convolution - [1407.6169v3] Multiplicative Complexity of Vector Valued Boolean Functions - [2511.02758v3] Finite free probability and  $S$  transforms of Jacobi processes - [2511.07739v3] A Lower Bound for the Fourier Entropy of Boolean Functions on the Biased Hypercube - [1806.02734v3] Spectral lower bounds for the orthogonal and projective ranks of a graph

## Empirical Test

- exit code: 0, elapsed: 94.45s

```
16, "conjecture_holds": true, "counterexample": "", "seed": 503}
TRIAL: {"metric_name": "free_mult_defect", "metric_value": 1.2174309166311343, "instances_tested": 1
TRIAL: {"metric_name": "free_mult_defect", "metric_value": 1.218306819204241, "instances_tested": 16
TRIAL: {"metric_name": "free_mult_defect", "metric_value": 1.2381510000250828, "instances_tested": 1
1 matrix: U < 0.05 log2(N) delta: n=3, U=0.0, delta=0.9908707865168539", "seed": 631}
TRIAL: {"metric_name": "free_mult_defect", "metric_value": 1.2508519087688272, "instances_tested": 1
TRIAL: {"metric_name": "free_mult_defect", "metric_value": 1.242301935238181, "instances_tested": 16
TRIAL: {"metric_name": "free_mult_defect", "metric_value": 1.2526578087928977, "instances_tested": 1
TRIAL: {"metric_name": "free_mult_defect", "metric_value": 1.219413383084314, "instances_tested": 16
TRIAL: {"metric_name": "free_mult_defect", "metric_value": 1.2409823554557478, "instances_tested": 1
TRIAL: {"metric_name": "free_mult_defect", "metric_value": 1.215808549927829, "instances_tested": 16
RESULT: FALSIFIED counterexample=Rank-1 matrix: U < 0.05 log2(N) delta: n=3, U=0.0, delta=0.7569444444
```

## Judge reasoning

Safety rail: critic\_challenge\_falsified | original: The test's RESULT line reports FALSIFIED with explicit counterexamples (Identity and Rank-1 matrices at n=3,4,5 having U=0 but  $\delta \approx 0.75-0.99$ ), violating | next: Restrict the conjecture to Boolean matrices of high combinatorial rank (e.g., exclude low-rank/structured M) or replace U with a lower bound on CC\_R rather than any certified upper bound.

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## Fekete Capacity of Laplacian Spectrum Upper-Bounds Max-Cut SoS-2 Gap

- **Verdict:** INCONCLUSIVE
- **Bridge:** Logarithmic potential theory — Fekete transfinite diameter / Vandermonde capacity of finite point sets on  $\mathbb{R}$ , in the Saff-Totik / Ransford / Hardin-Saff tradition (equilibrium measures, Robin constants, weighted log-energy). This corner of classical analysis has essentially no presence in SoS / CSP integrality-gap literature (arXiv 'transfinite diameter' AND ('sum of squares' OR 'Lasserre' OR 'Max-Cut') returns 0 direct hits, with <5 adjacent random-matrix papers on Wigner-type spectral statistics). × Degree-2 Sum-of-Squares / Lasserre integrality gap for Max-Cut on small graphs, accessed through the Delorme-Poljak eigenvalue upper bound  $\text{SDP}_2(G) \leq (n/4) \cdot \lambda_n(L_G)$  (which coincides with degree-2 SoS for vertex-transitive graphs and is a valid SoS-2 upper bound in general). This is the Goemans-Williamson / Schoenebeck-Khot-Vishnoi / Barak-Steurer regime targeted by SOS\_HIERARCHY.
- **Recorded:** 2026-05-17 20:47 UTC
- **Entry ID:** 999ba4b45fab

## Statement

For a connected graph  $G$  on  $n$  vertices with Laplacian  $L_G$  having nontrivial eigenvalues  $0 < \lambda_2 \leq \dots \leq \lambda_n$ , let  $\tilde{\sigma}(G) = \{\lambda_2/\lambda_n, \dots, \lambda_{n-1}/\lambda_n, 1\}$  be the spectrum rescaled into  $(0,1]$ , and define the normalized Fekete capacity  $\text{Cap}(G) := (\prod_{2 \leq i < j \leq n} |\tilde{\lambda}_i - \tilde{\lambda}_j|)^{2/((n-1)(n-2))} \in [0,1]$  (the transfinite diameter of the rescaled spectrum, taken as 0 when any pair coincides exactly). Let  $\text{gap}(G) := n \cdot \lambda_n / (4 \cdot \text{MaxCut}(G)) - 1 \geq 0$  be the Delorme-Poljak / SoS-2 multiplicative integrality gap. Conjecture: there is an absolute constant  $C \leq 1$  such that for every connected graph  $G$  on  $6 \leq n \leq 16$  vertices,  $\text{gap}(G) \leq C \cdot \text{Cap}(G)$ ; equivalently, any nonzero SoS-2 integrality gap forces the rescaled Laplacian spectrum to have positive transfinite diameter, with the gap controlled linearly by it. A single graph  $G$  with  $\text{gap}(G) > C \cdot \text{Cap}(G)$  by additive margin  $> 0.01$  refutes the conjecture.

## Rationale

Bipartite-like graphs ( $K_{\{n/2, n/2\}}$ , balanced complete multipartite) saturate the Delorme–Poljak bound ( $\text{gap} = 0$ ) AND have multiply-repeated Laplacian eigenvalues ( $\text{Cap} = 0$ ), while expander 3-regular graphs have a well-spread spectrum ( $\text{Cap} > 0$ ) AND a positive gap ( $\approx 0.087$ ); the conjecture quantifies this co-occurrence by bounding the SoS-2 gap above by the geometric-mean spectral-gap product, a classical potential-theoretic invariant from Saff–Totik that has never been used as a hardness functional. The invariant is computed on the *graph* (a structural object) — not on a truth table — so it is shielded from NATURAL\_PROOFS; it uses logarithms (non-ring), so it is shielded from ALGEBRIZATION; and it concerns concrete graphs, not oracles, so it is shielded from RELATIVIZATION. The bridge sharpens existing Delorme–Poljak / Goemans–Williamson analysis by replacing the single  $\lambda_n$  with the full Vandermonde-style multiplicative structure of  $\sigma(L_G)$ .

## Novelty

- Judge: NOVEL over 9 arXiv hits

Top hits consulted: - [1107.1630v3] On the Integrality Gap of the Subtour LP for the 1,2-TSP - [1207.5696v1] Beyond Max-Cut:  $\lambda$ -Extendible Properties Parameterized Above the Poljak–Turzík Bound - [2205.10968v4] Eigenvalue bounds of the Kirchhoff Laplacian - [1210.8059v3] Equidistribution estimates for Fekete points on complex manifolds - [1301.6896v1] Laplacians on periodic discrete graphs

## Empirical Test

- exit code: 0, elapsed: 0.15s

```
6, "conjecture_holds": false, "counterexample": "n=8, gap=0.228280467467467, cap=0.1775420094450226,
TRIAL: {"metric_name": "gap_minus_cap", "metric_value": 0.0, "metric_std": 0.0, "instances_tested": 1}
TRIAL: {"metric_name": "gap_minus_cap", "metric_value": 0.0, "metric_std": 0.0, "instances_tested": 1}
TRIAL: {"metric_name": "gap_minus_cap", "metric_value": 0.0, "metric_std": 0.0, "instances_tested": 1}
TRIAL: {"metric_name": "gap_minus_cap", "metric_value": 0.0, "metric_std": 0.0, "instances_tested": 1}
TRIAL: {"metric_name": "gap_minus_cap", "metric_value": 0.0, "metric_std": 0.0, "instances_tested": 1}
```

## Judge reasoning

Safety rail: critic\_challenge\_falsified | original: Multiple instances violate  $\text{gap}(G) \leq \text{Cap}(G) + 0.01$  by large margins; e.g.,  $n=8$  with  $\text{gap} \approx 0.2587$ ,  $\text{Cap} \approx 0.1688$  (margin  $\approx 0.09$ ), and degenerate cases like se | next: Replace Fekete capacity with a degeneracy-robust separation surrogate (e.g., min pairwise gap regularized as  $\log(1/(\epsilon + \min_i |\tilde{\lambda}_i - \tilde{\lambda}_j|))$ ) or restrict to graphs with simple Laplacian spectra before re-testing the gap bound.

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## Polymer Cluster-Expansion Convergence Radius Bounds Tree-Res Size

- **Verdict:** INCONCLUSIVE
- **Bridge:** Statistical mechanics — Fernandez–Procacci / Kotecký–Preiss cluster-expansion convergence radius for abstract polymer models (Mayer 1937; Kotecký–Preiss 1986; Fernandez–Procacci 2007; Bissacot–Fernandez–Procacci 2010). An analytic structural invariant of finite graphs essentially absent from proof complexity outside loose Lovász–Local-Lemma analogies; an arXiv search ‘cluster expansion’ AND (‘Resolution refutation’ OR ‘DPLL’ OR ‘Frege’ OR ‘proof complexity’) returns no direct hits and  $<5$  adjacent papers, none using FP fixed-point activities as a CNF invariant.  $\times$  Tree-like Resolution / DPLL refutation tree size  $t(F)$  for unsatisfiable 3-CNFs, in the Ben-Sasson–Wigderson–Krajíček bounded-arithmetic regime where  $\neg F$  is  $\Sigma^b_0$  and  $\log_2 t(F)$  controls  $V^0$ -witnessing complexity — a Frege/Extended-Frege stepping stone via the Pudlák–Buss feasible-interpolation and Krajíček EF-depth simulation chain.
- **Recorded:** 2026-05-17 21:20 UTC

- **Entry ID:** 3905f361f983

## Statement

For an unsatisfiable 3-CNF  $F$  on  $n$  variables with clauses  $C_1, \dots, C_m$ , build the clause-overlap graph  $G(F)$  (vertices = clauses; edge iff the two clauses share a variable) and let  $\mu(F) \in (0,1]^m$  be the unique fixed point of the Fernandez-Procacci iteration  $\mu_v \leftarrow 1/(1 + \sum_{u \sim v} \mu_u)$  initialised at  $\mu_v(0)=1$ ; define the polymer convergence defect  $\Lambda(F) := -\sum_v \log_2 \mu_v$ . We conjecture there is an absolute constant  $c \geq 1/8$  such that  $\log_2 t(F) \geq c \cdot \Lambda(F)$  for every unsatisfiable 3-CNF  $F$ , with the bound near-tight on random 3-CNFs at  $\alpha \approx 4.26$ . A single unsatisfiable  $F$  with  $\log_2 t(F) < (1/8) \cdot \Lambda(F)$  falsifies the conjecture.

## Rationale

Cluster-expansion convergence is the analytic dual of clause-overlap expansion: the FP activities  $\mu_v$  contract precisely when local clause interactions admit a convergent Mayer series, mirroring the same expansion mechanism underlying Ben-Sasson-Wigderson width lower bounds — but packaged as a single-number invariant of the SYNTACTIC formula, not of any truth-table, so it dodges the Razborov-Rudich naturalization route. The construction is non-arithmetizing (the fixed point is a graph-combinatorial invariant of the CNF symbol level, not a polynomial extension of the Boolean ring), avoiding the algebrization wall hit by GCT cubic-form attempts. If correct,  $\Lambda(F)$  would supply a physics-style ‘free-energy’ depth measure for DPLL and, via interpolation, the first quantitative structural bridge from cavity-method analysis to Frege/EF lower bounds.

## Novelty

- Judge: NOVEL over 9 arXiv hits

Top hits consulted: - [1705.04385v2] A correction to a remark in a paper by Procacci and Yuhjtman: new lower bounds for the convergence radius of the virial - [0605041v2] Cluster expansion for abstract polymer models. New bounds from an old approach - [1002.3261v1] On the convergence of cluster expansions for polymer gases - [1904.03250v1] Velocity jump in the crack propagation induced on a semi-crystalline polymer sheet by constant-speed stretching - [0906.0693v3] An improved lower bound on the counterfeit coins problem

## Empirical Test

- exit code: 0, elapsed: 0.35s

```
es_tested': 0, 'conjecture_holds': False, 'counterexample': 'No valid instances generated'}
TRIAL: {'metric_name': 'r(F)', 'metric_value': 0.0, 'instances_tested': 0, 'conjecture_holds': False}
TRIAL: {'metric_name': 'r(F)', 'metric_value': 0.0, 'instances_tested': 0, 'conjecture_holds': False}
TRIAL: {'metric_name': 'r(F)', 'metric_value': 0.0, 'instances_tested': 0, 'conjecture_holds': False}
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TRIAL: {'metric_name': 'r(F)', 'metric_value': 0.0, 'instances_tested': 0, 'conjecture_holds': False}
TRIAL: {'metric_name': 'r(F)', 'metric_value': 0.0, 'instances_tested': 0, 'conjecture_holds': False}
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TRIAL: {'metric_name': 'r(F)', 'metric_value': 0.0, 'instances_tested': 0, 'conjecture_holds': False}
TRIAL: {'metric_name': 'r(F)', 'metric_value': 0.0, 'instances_tested': 0, 'conjecture_holds': False}
TRIAL: {'metric_name': 'r(F)', 'metric_value': 0.0, 'instances_tested': 0, 'conjecture_holds': False}
TRIAL: {'metric_name': 'r(F)', 'metric_value': 0.0, 'instances_tested': 0, 'conjecture_holds': False}
TRIAL: {'metric_name': 'r(F)', 'metric_value': 0.0, 'instances_tested': 0, 'conjecture_holds': False}
TRIAL: {'metric_name': 'r(F)', 'metric_value': 0.0, 'instances_tested': 0, 'conjecture_holds': False}
TRIAL: {'metric_name': 'r(F)', 'metric_value': 0.0, 'instances_tested': 0, 'conjecture_holds': False}
RESULT: FALSIFIED counterexample=No valid instances generated first_failing_seed=11
```

## Judge reasoning

The test generated zero valid instances across all 210 trials (instances\_tested: 0, metric\_value: 0.0), so the conjecture was never empirically evaluated. The ‘FALSIFIED’ RESULT line reflects a gener-

ator/construction failure, not a genuine counterexample to the bound. | next: Fix the unsatisfiable 3-CNF generator (e.g., sample random 3-CNFs at  $\alpha=4.26$  and filter via a SAT solver, or use known unsat constructions like PHP<sub>n</sub>+1<sup>n</sup> encodings) so that  $t^*(F)$  and  $\Lambda(F)$  can actually be computed for all 210